CHAPTER 6

An Overview of Organic Reactions



FIGURE 6.1 Many chemical reactions are like pole vaulters going over the bar. They need a big, initial push of activation energy. (credit: modification of work "UChicago Pole Vault" by Eric Guo/Flickr, CC BY 2.0)

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WHY THIS CHAPTER? All chemical reactions, whether they take place in the laboratory or in living organisms, follow the same "rules." Reactions in living organisms often look more complex than laboratory reactions because of the size of the biomolecules and the involvement of biological catalysts called enzymes, but the principles governing all chemical reactions are the same.

To understand both organic and biological chemistry, it's necessary to know not just *what* occurs but also *why* and *how* chemical reactions take place. In this chapter, we'll start with an overview of the fundamental kinds of organic reactions, we'll see why reactions occur, and we'll see how reactions can be described. Once this

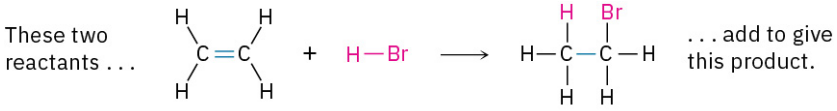
background is out of the way, we'll then be ready to begin studying the details of organic chemistry in future chapters.

When first approached, organic chemistry might seem overwhelming. It's not so much that any one part is difficult to understand, it's that there are so many parts: tens of millions of compounds, dozens of functional groups, and an apparently endless number of reactions. With study, though, it becomes evident that there are only a few fundamental ideas that underlie all organic reactions. Far from being a collection of isolated facts, organic chemistry is a beautifully logical subject that is unified by a few broad themes. When these themes are understood, learning organic chemistry becomes much easier and memorization is minimized. The aim of this book is to describe the themes and clarify the patterns that unify organic chemistry in future chapters.

6.1 Kinds of Organic Reactions

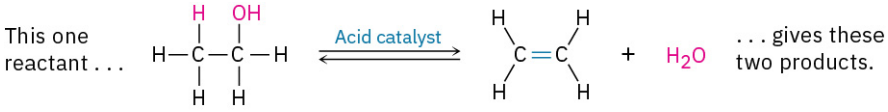
Organic chemical reactions can be organized broadly in two ways—by what kinds of reactions occur and by how those reactions occur. Let's look first at the kinds of reactions that take place. There are four general types of organic reactions: *additions, eliminations, substitutions,* and *rearrangements*.

- **Addition reactions** occur when two reactants add together to form a single product with no atoms "left over." An example that we'll be studying soon is the reaction of an alkene, such as ethylene, with HBr to yield an alkyl bromide.



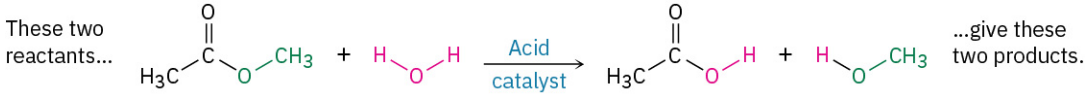
Ethylene (an alkene) **Bromoethane (an alkyl halide)**

- **Elimination reactions** are, in a sense, the opposite of addition reactions. They occur when a single reactant splits into two products, often with the formation of a small molecule such as water or HBr. An example is the acid-catalyzed reaction of an alcohol to yield water and an alkene.



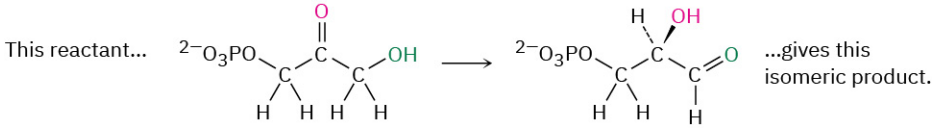
Ethanol (an alcohol) **Ethylene (an alkene)**

- **Substitution reactions** occur when two reactants exchange parts to give two new products. An example is the reaction of an ester such as methyl acetate with water to yield a carboxylic acid plus an alcohol. Similar reactions occur in many biological pathways, including the metabolism of dietary fats.



Methyl acetate (an ester) **Acetic acid (a carboxylic acid)** **Methanol (an alcohol)**

- **Rearrangement reactions** occur when a single reactant undergoes a reorganization of bonds and atoms to yield an isomeric product. An example is the conversion of dihydroxyacetone phosphate into its constitutional isomer glyceraldehyde 3-phosphate, a step in the glycolysis pathway by which carbohydrates are metabolized.



Dihydroxyacetone phosphate **Glyceraldehyde 3-phosphate**

PROBLEM Classify each of the following reactions as an addition, elimination, substitution, or rearrangement:

6-1 **(a)** $CH_3Br + KOH \longrightarrow CH_3OH + KBr$ **(b)** $CH_3CH_2Br \longrightarrow H_2C{=}CH_2 + HBr$

(c) $H_2C{=}CH_2 + H_2 \longrightarrow CH_3CH_3$

6.2 How Organic Reactions Occur: Mechanisms

Having looked at the kinds of reactions that take place, let's now see how they occur. An overall description of how a reaction occurs is called a **reaction mechanism**. A mechanism describes in detail exactly what takes place at each stage of a chemical transformation—which bonds are broken and in what order, which bonds are formed and in what order, and what the relative rates are for each step. A complete mechanism must also account for all reactants used and all products formed.

All chemical reactions involve bond-breaking and bond-making. When two molecules come together, react, and yield products, specific bonds in the reactant molecules are broken and specific bonds in the product molecules are formed. Fundamentally, there are two ways in which a covalent two-electron bond can break. A bond can break in an electronically unsymmetrical way so that both bonding electrons remain with one product fragment, leaving the other with a vacant orbital, or a bond can break in an electronically symmetrical way so that one electron remains with each product fragment. The unsymmetrical cleavage is said to be *heterolytic*, and the symmetrical cleavage is said to be *homolytic*.

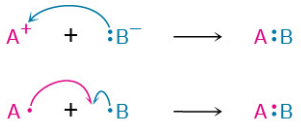
We'll develop this point in more detail later, but note for now that the movement of *two* electrons in the unsymmetrical process is indicated using a full-headed curved arrow (↷), whereas the movement of *one* electron in the symmetrical process is indicated using a half-headed, or "fishhook," arrow (↶).



Unsymmetrical bond-breaking (polar):
two bonding electrons stay with one product.

Symmetrical bond-breaking (radical):
one bonding electron stays with each product.

Just as there are two ways in which a bond can break, there are two ways in which a covalent two-electron bond can form. A bond can form in an electronically unsymmetrical way if both bonding electrons are donated to the new bond by one reactant, or in a symmetrical way if one electron is donated by each reactant.



Unsymmetrical bond-making (polar):
two bonding electrons are donated by one reactant.

Symmetrical bond-making (radical):
one bonding electron is donated by each reactant.

Processes that involve unsymmetrical bond-breaking and bond-making are called **polar reactions**. Polar reactions involve species that have an even number of electrons and thus have only electron pairs in their orbitals. Polar processes are by far the more common reaction type in both organic and biological chemistry, and a large part of this book is devoted to their description.

Processes that involve symmetrical bond-breaking and bond-making are called **radical reactions**. A **radical**, often called a **free radical**, is a neutral chemical species that contains an odd number of electrons and thus has a single, unpaired electron in one of its orbitals.

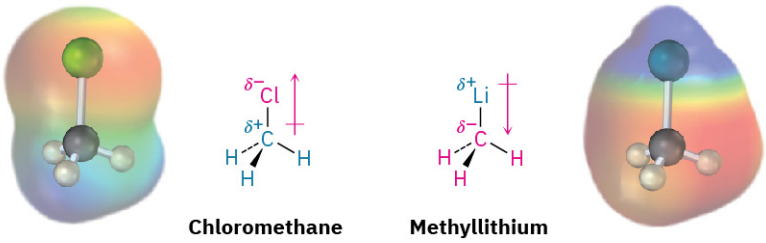
In addition to polar and radical reactions, there is a third, less commonly encountered process called a *pericyclic reaction*. Rather than explain pericyclic reactions now, though, we'll look at them more carefully in Chapter 30.

6.3 Polar Reactions

Polar reactions occur because of the electrical attraction between positively polarized and negatively polarized centers on functional groups in molecules. To see how these reactions take place, let's first recall the discussion of polar covalent bonds in **Section 2.1** and then look more deeply into the effects of bond polarity on organic molecules.

Most organic compounds are electrically neutral; they have no net charge, either positive or negative. We saw in **Section 2.1**, however, that certain bonds within a molecule, particularly the bonds in functional groups, are polar. Bond polarity is a consequence of an unsymmetrical electron distribution in a bond and is due to the difference in electronegativity of the bonded atoms.

Elements such as oxygen, nitrogen, fluorine, and chlorine are more electronegative than carbon, so a carbon atom bonded to one of these atoms has a partial positive charge ($\delta+$). Metals are less electronegative than carbon, so a carbon atom bonded to a metal has a partial negative charge ($\delta-$). Electrostatic potential maps of chloromethane and methyllithium illustrate these charge distributions, showing that the carbon atom in chloromethane is electron-poor (blue) while the carbon in methyllithium is electron-rich (red).



Chloromethane **Methyllithium**

The polarity patterns of some common functional groups are shown in **TABLE 6.1**. Note that carbon is always positively polarized except when bonded to a metal.

TABLE 6.1 Polarity Patterns in Some Common Functional Groups

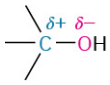
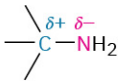
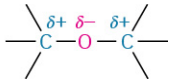
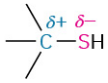
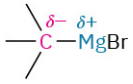
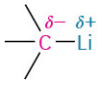
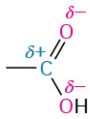
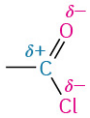
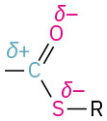
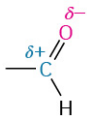
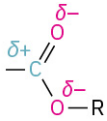
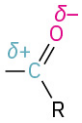
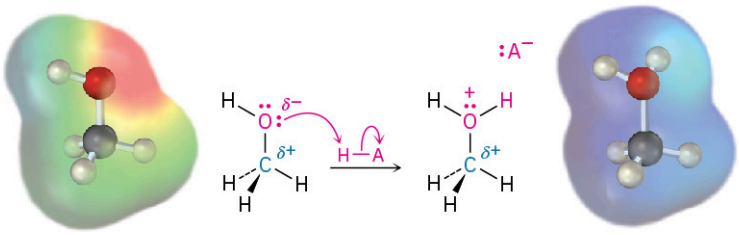
Compound type	Functional group structure
Alcohol	
Alkene	 Symmetrical, nonpolar
Alkyl halide	
Amine	
Ether	
Thiol	
Nitrile	
Grignard reagent	
Alkyllithium	

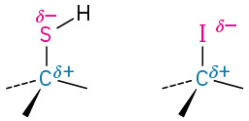
TABLE 6.1 Polarity Patterns in Some Common Functional Groups

Compound type	Functional group structure
Carbonyl	
Carboxylic acid	
Carboxylic acid chloride	
Thioester	
Aldehyde	
Ester	
Ketone	

This discussion of bond polarity is oversimplified in that we've considered only bonds that are inherently polar due to differences in electronegativity. Polar bonds can also result from the interaction of functional groups with acids or bases. Take an alcohol such as methanol, for example. In neutral methanol, the carbon atom is somewhat electron-poor because the electronegative oxygen attracts the electrons in the C–O bond. On protonation of the methanol oxygen by an acid, however, a full positive charge on oxygen attracts the electrons in the C–O bond much more strongly and makes the carbon much more electron-poor. We'll see numerous examples throughout this book of reactions that are catalyzed by acids because of the resultant increase in bond polarity upon protonation.

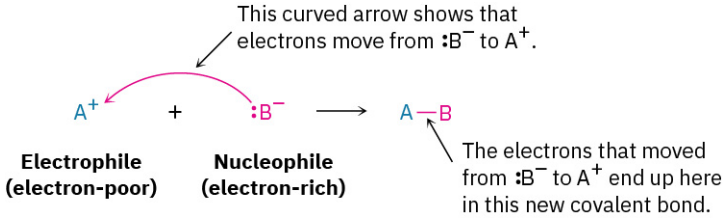


Yet a further consideration is the polarizability (as opposed to polarity) of atoms in a molecule. As the electric field around a given atom changes because of changing interactions with solvent or other polar molecules nearby, the electron distribution around that atom also changes. The measure of this response to an external electrical influence is called the **polarizability** of the atom. Larger atoms with more loosely held electrons are more polarizable, and smaller atoms with fewer, tightly held electrons are less polarizable. Thus, sulfur is more polarizable than oxygen, and iodine is more polarizable than chlorine. The effect of this higher polarizability of sulfur and iodine is that carbon–sulfur and carbon–iodine bonds, although nonpolar according to electronegativity values (FIGURE 2.3), nevertheless usually react as if they were polar.



What does functional-group polarity mean with respect to chemical reactivity? Because unlike charges attract, the fundamental characteristic of all polar organic reactions is that electron-rich sites react with electron-poor sites. Bonds are made when an electron-rich atom donates a pair of electrons to an electron-poor atom, and bonds are broken when one atom leaves with both electrons from the former bond.

As we saw in Section 2.11, the movement of an electron pair during a polar reaction is indicated using a curved, full-headed arrow to show where electrons move when reactant bonds are broken and product bonds are formed during the reaction.



In referring to the electron-rich and electron-poor species involved in polar reactions, chemists use the words *nucleophile* and *electrophile*. A **nucleophile** is a substance that is "nucleus-loving." (Remember that a nucleus is positively charged.) A nucleophile has a negatively polarized, electron-rich atom and can form a bond by donating a pair of electrons to a positively polarized, electron-poor atom. Nucleophiles can be either neutral or negatively charged; ammonia, water, hydroxide ion, and chloride ion are examples. An **electrophile**, by contrast, is "electron-loving." An electrophile has a positively polarized, electron-poor atom and can form a bond by accepting a pair of electrons from a nucleophile. Electrophiles can be either neutral or positively charged. Acids (H^+ donors), alkyl halides, and carbonyl compounds are examples (FIGURE 6.2).

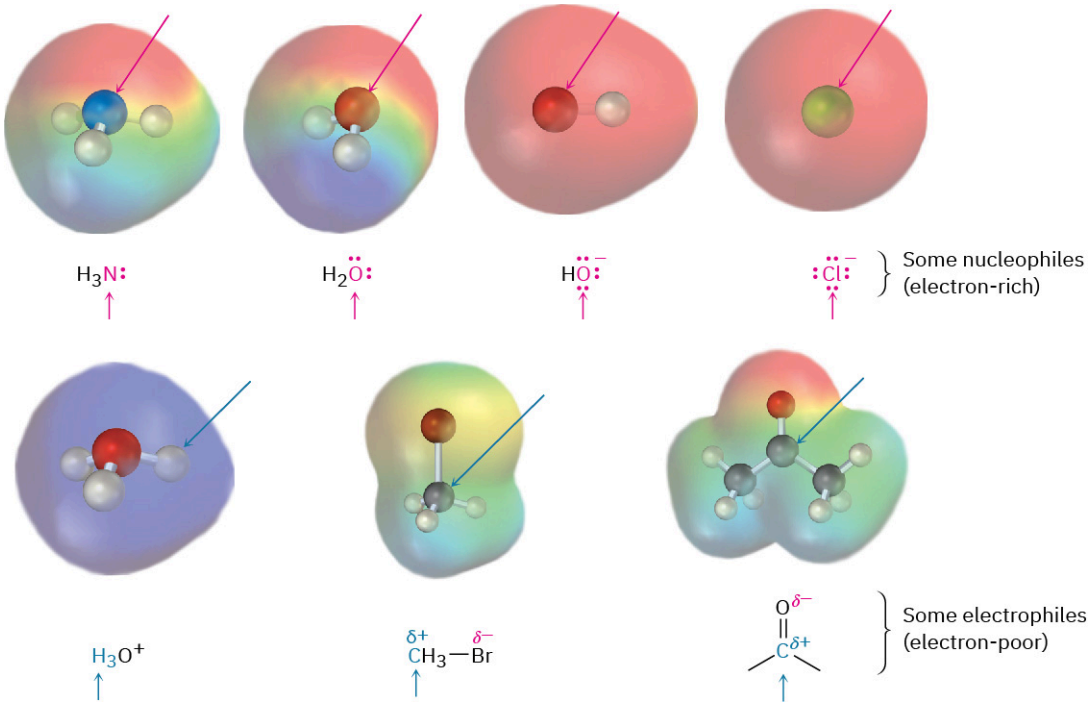


FIGURE 6.2 Some nucleophiles and electrophiles. Electrostatic potential maps identify the nucleophilic **(negative)** and electrophilic **(positive)** atoms.

Note that neutral compounds can often react either as nucleophiles or as electrophiles, depending on the circumstances. After all, if a compound is neutral yet has an electron-*rich* nucleophilic site, it must also have a corresponding electron-*poor* electrophilic site. Water, for instance, acts as an electrophile when it donates H^+ but acts as a nucleophile when it donates a nonbonding pair of electrons. Similarly, a carbonyl compound acts as an electrophile when it reacts at its positively polarized carbon atom, yet acts as a nucleophile when it reacts at its negatively polarized oxygen atom.

If the definitions of nucleophiles and electrophiles sound similar to those given in **Section 2.11** for Lewis acids and Lewis bases, that's because there is indeed a correlation. Lewis bases are electron donors and behave as nucleophiles, whereas Lewis acids are electron acceptors and behave as electrophiles. Thus, much of organic chemistry is explainable in terms of acid–base reactions. The main difference is that the words *acid* and *base* are used broadly in all fields of chemistry, while the words *nucleophile* and *electrophile* are used primarily in organic chemistry when carbon bonding is involved.

WORKED EXAMPLE 6.1

Identifying Electrophiles and Nucleophiles

Which of the following species is likely to behave as a nucleophile and which as an electrophile?

(a) NO_2^+ **(b)** CN^- **(c)** CH_3NH_2 **(d)** $(CH_3)_3S^+$

Strategy

A nucleophile has an electron-rich site, either because it is negatively charged or because it has a functional group containing an atom that has a lone pair of electrons. An electrophile has an electron-poor site, either because it is positively charged or because it has a functional group containing an atom that is positively polarized.

Solution

(a) NO_2^+ (nitronium ion) is likely to be an electrophile because it is positively charged.

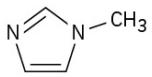
(b) $:C{\equiv}N^-$ (cyanide ion) is likely to be a nucleophile because it is negatively charged.

(c) CH_3NH_2 (methylamine) might be either a nucleophile or an electrophile, depending on the circumstances. The lone pair of electrons on the nitrogen atom makes methylamine a potential nucleophile, while positively polarized N–H hydrogens make methylamine a potential acid (electrophile).

(d) $(CH_3)_3S^+$ (trimethylsulfonium ion) is likely to be an electrophile because it is positively charged.

PROBLEM 6-2 Which of the following species are likely to be nucleophiles and which electrophiles? Which might be both?

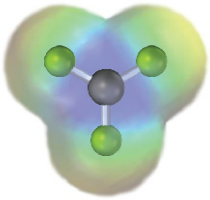
(a) CH_3Cl **(b)** CH_3S^- **(c)**



(d)



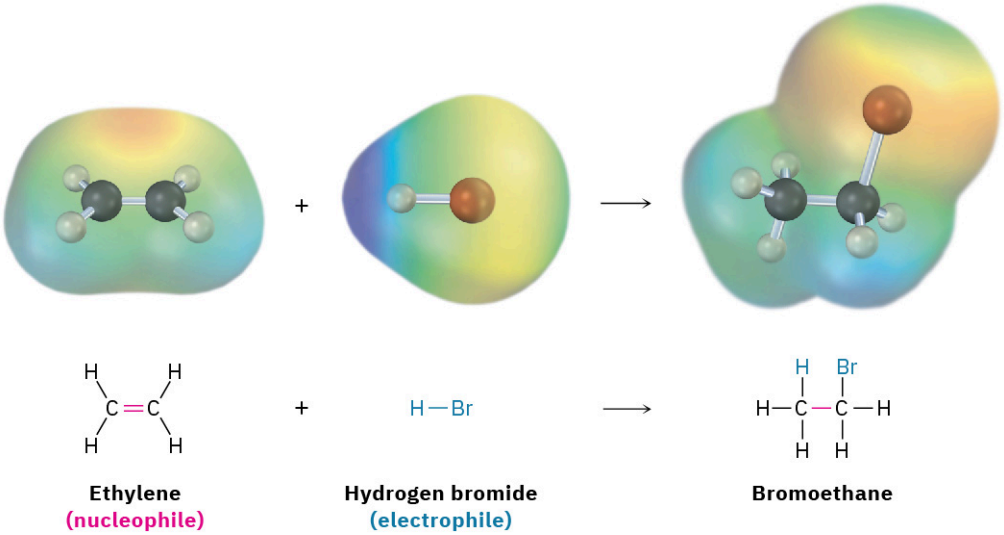
PROBLEM 6-3 An electrostatic potential map of boron trifluoride is shown. Is BF_3 likely to be a nucleophile or an electrophile? Draw a Lewis structure for BF_3, and explain your answer.



BF_3

6.4 An Example of a Polar Reaction: Addition of HBr to Ethylene

Let's look at a typical polar process—the addition reaction of an alkene, such as ethylene, with hydrogen bromide. When ethylene is treated with HBr at room temperature, bromoethane is produced. Overall, the reaction can be formulated as



The reaction is an example of a polar reaction type known as an *electrophilic addition reaction* and can be understood using the general ideas discussed in the previous section. Let's begin by looking at the two reactants.

What do we know about ethylene? We know from **Section 1.8** that a carbon–carbon double bond results from the orbital overlap of two sp^2-hybridized carbon atoms. The σ part of the double bond results from sp^2–sp^2 overlap, and the π part results from p–p overlap.

What kind of chemical reactivity might we expect from a C═C bond? We know that alkanes, such as ethane, are relatively inert because all valence electrons are tied up in strong, nonpolar, C–C and C–H bonds. Furthermore, the bonding electrons in alkanes are relatively inaccessible to approaching reactants because they are sheltered in σ bonds between nuclei. The electronic situation in alkenes is quite different, however. For one thing, double

bonds have a greater electron density than single bonds—four electrons in a double bond versus only two in a single bond. In addition, the electrons in the π bond are accessible to approaching reactants because they are located above and below the plane of the double bond rather than being sheltered between the nuclei (**FIGURE 6.3**). As a result, the double bond is nucleophilic and the chemistry of alkenes is dominated by reactions with electrophiles.

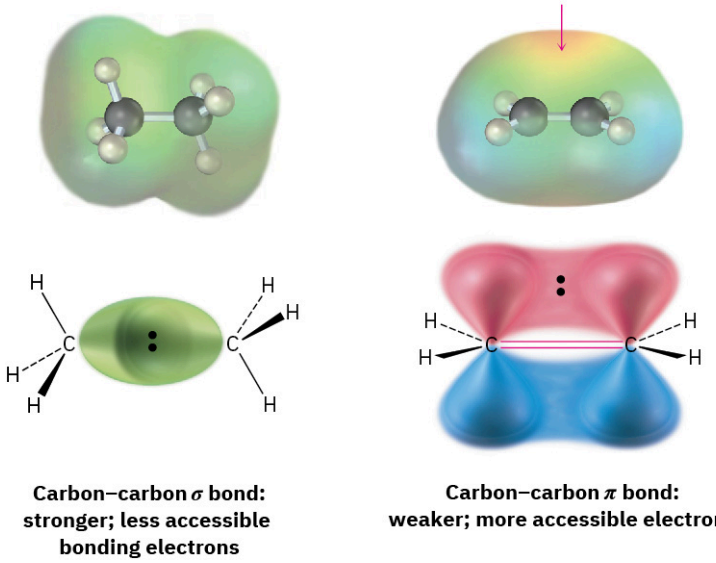


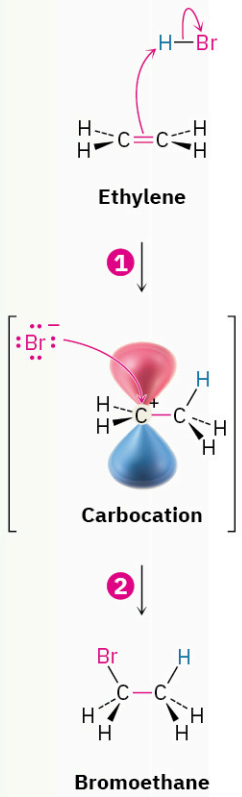
FIGURE 6.3 A comparison of carbon–carbon single and double bonds. A double bond is both more accessible to approaching reactants than a single bond and more electron-rich (more nucleophilic). An electrostatic potential map of ethylene indicates that the double bond is the region of **highest negative charge**.

What about the second reactant, HBr? As a strong acid, HBr is a powerful proton (H^+) donor and electrophile. Thus, the reaction between HBr and ethylene is a typical electrophile–nucleophile combination, characteristic of all polar reactions.

We'll see more details about alkene electrophilic addition reactions shortly, but for the present we can imagine the reaction as taking place by the pathway shown in **FIGURE 6.4**. The reaction begins when the alkene nucleophile donates a pair of electrons from its C=C bond to HBr to form a new C–H bond plus Br^-, as indicated by the path of the curved arrows in the first step of **FIGURE 6.4**. One curved arrow begins at the middle of the double bond (the source of the electron pair) and points to the hydrogen atom in HBr (the atom to which a bond will form). This arrow indicates that a new C–H bond forms using electrons from the former C=C bond. Simultaneously, a second curved arrow begins in the middle of the H–Br bond and points to the Br, indicating that the H–Br bond breaks and the electrons remain with the Br atom, giving Br^-.

FIGURE 6.4 MECHANISM

The electrophilic addition reaction of ethylene and HBr. The reaction takes place in two steps, both of which involve electrophile–nucleophile interactions.



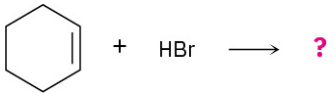
1. A hydrogen atom on the electrophile HBr is attacked by π electrons from the nucleophilic double bond, forming a new C–H bond. This leaves the other carbon atom with a + charge and a vacant *p* orbital. Simultaneously, two electrons from the H–Br bond move onto bromine, giving bromide anion.

2. Bromide ion donates an electron pair to the positively charged carbon atom, forming a C–Br bond and yielding the neutral addition product.

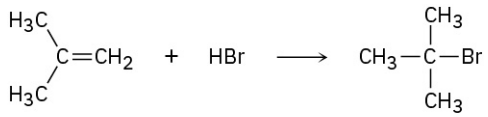
When one of the alkene carbon atoms bonds to the incoming hydrogen, the other carbon atom, having lost its share of the double-bond electrons, now has only six valence electrons and is left with a positive charge. This positively charged species—a carbon-cation, or **carbocation**—is itself an electrophile that can accept an electron pair from nucleophilic Br^- anion in a second step, forming a C–Br bond and yielding the observed addition product. Once again, a curved arrow in FIGURE 6.4 shows the electron-pair movement from Br^- to the positively charged carbon.

The electrophilic addition of HBr to ethylene is only one example of a polar process; there are many others that we'll study in depth in later chapters. But regardless of the details of individual reactions, all polar reactions take place between an electron-poor site and an electron-rich site and involve the donation of an electron pair from a nucleophile to an electrophile.

PROBLEM 6-4 What product would you expect from reaction of cyclohexene with HBr? With HCl?



PROBLEM 6-5 Reaction of HBr with 2-methylpropene yields 2-bromo-2-methylpropane. What is the structure of the carbocation formed during the reaction? Show the mechanism of the reaction.



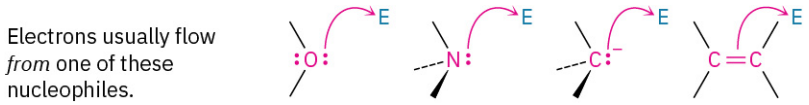
2-Methylpropene **2-Bromo-2-methylpropane**

6.5 Using Curved Arrows in Polar Reaction Mechanisms

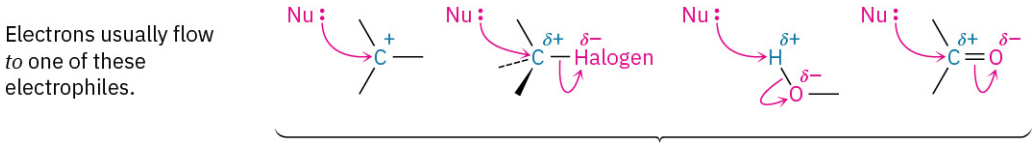
It takes practice to use curved arrows properly in reaction mechanisms, but there are a few rules and a few common patterns you should look for that will help you become more proficient:

RULE 1

Electrons move *from* a nucleophilic source (Nu: or Nu:$^-$) *to* an electrophilic sink (E or E$^+$). The nucleophilic source must have an electron pair available, usually either as a lone pair or in a multiple bond. For example:

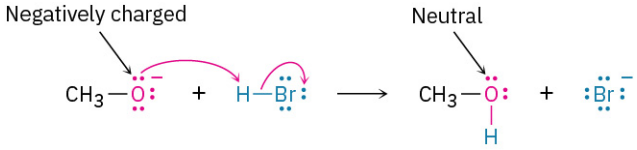


The electrophilic sink must be able to accept an electron pair, usually because it has either a positively charged atom or a positively polarized atom in a functional group. For example:

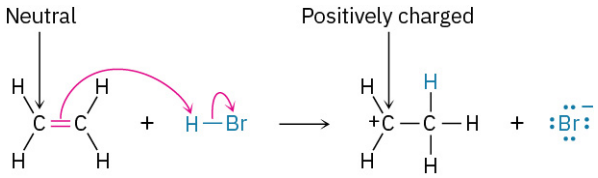


RULE 2

The nucleophile can be either negatively charged or neutral. If the nucleophile is negatively charged, the atom that donates an electron pair becomes neutral. For example:

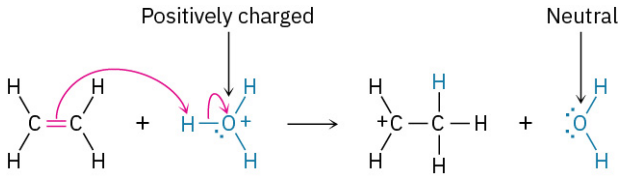


If the nucleophile is neutral, the atom that donates the electron pair acquires a positive charge. For example:

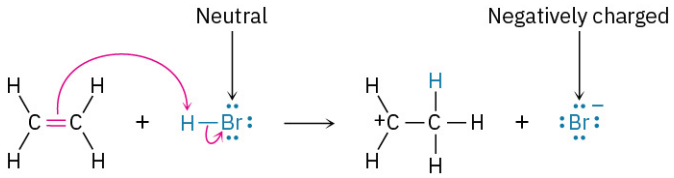


RULE 3

The electrophile can be either positively charged or neutral. If the electrophile is positively charged, the atom bearing that charge becomes neutral after accepting an electron pair. For example:



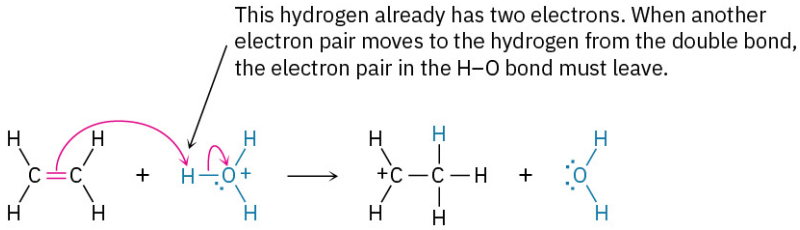
If the electrophile is neutral, the atom that ultimately accepts the electron pair acquires a negative charge. For this to happen, however, the negative charge must be stabilized by being on an electronegative atom such as oxygen, nitrogen, or a halogen. Carbon and hydrogen do not typically stabilize a negative charge. For example:



The result of Rules 2 and 3 together is that charge is conserved during the reaction. A negative charge in one of the reactants gives a negative charge in one of the products, and a positive charge in one of the reactants gives a positive charge in one of the products.

RULE 4

The octet rule must be followed. That is, no second-row atom can be left with ten electrons (or four for hydrogen). If an electron pair moves *to* an atom that already has an octet (or two electrons for hydrogen), another electron pair must simultaneously move from that atom to maintain the octet. When two electrons move from the C=C bond of ethylene to the hydrogen atom of H_3O^+, for instance, two electrons must leave that hydrogen. This means that the H–O bond must break and the electrons must stay with the oxygen, giving neutral water.

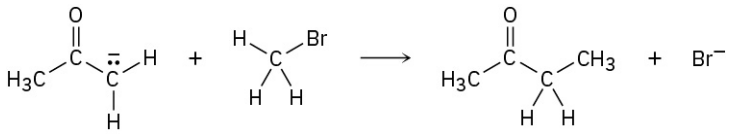


Worked Example 6.2 gives another example of drawing curved arrows.

WORKED EXAMPLE 6.2

Using Curved Arrows in Reaction Mechanisms

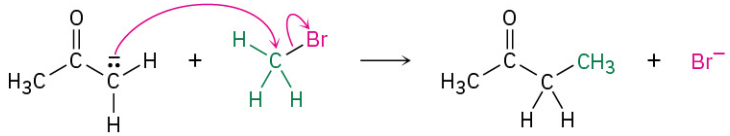
Add curved arrows to the following polar reaction to show the flow of electrons:



Strategy

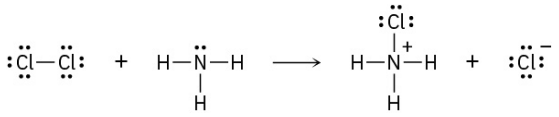
Look at the reaction, and identify the bonding changes that have occurred. In this case, a C–Br bond has broken and a C–C bond has formed. The formation of the C–C bond involves donation of an electron pair from the nucleophilic carbon atom of the reactant on the left to the electrophilic carbon atom of CH_3Br, so we draw a curved arrow originating from the lone pair on the negatively charged C atom and pointing to the C atom of CH_3Br. At the same time that the C–C bond forms, the C–Br bond must break so that the octet rule is not violated. We therefore draw a second curved arrow from the C–Br bond to Br. The bromine is now a stable Br^- ion.

Solution

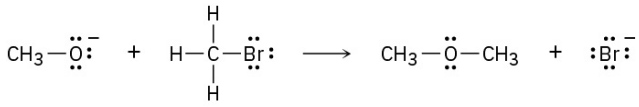


PROBLEM 6-6 Add curved arrows to the following polar reactions to indicate the flow of electrons in each:

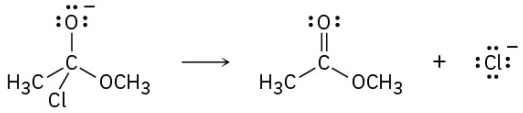
(a)



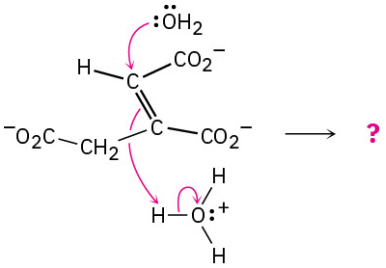
(b)



(c)



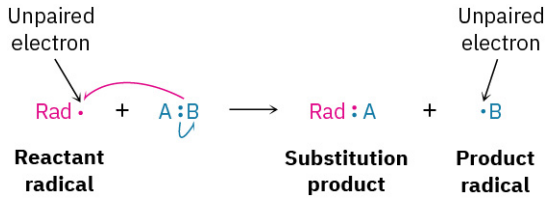
PROBLEM 6-7 Predict the products of the following polar reaction, a step in the citric acid cycle for food metabolism, by interpreting the flow of electrons indicated by the curved arrows:



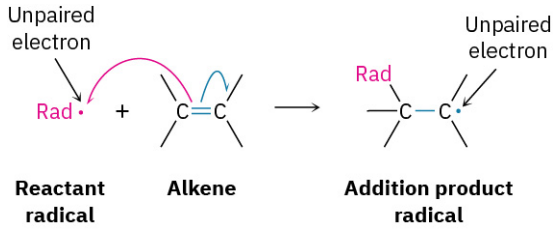
6.6 Radical Reactions

Radical reactions are much less common than polar reactions but are nevertheless important in some industrial processes and biological pathways. We'll look at them in more detail in **Sections 10.2** and **10.3** but will briefly see how they occur at this point.

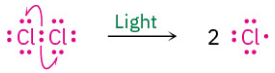
A radical is highly reactive because it contains an atom with an odd number of electrons (usually seven) in its valence shell, rather than a noble-gas octet. The radical can achieve a valence-shell octet in several ways however. For instance, it might abstract an atom and one bonding electron from another reactant, leaving behind a new radical. The net result is a radical substitution reaction.



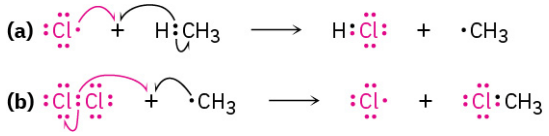
Alternatively, a reactant radical might add to a double bond, taking one electron from the double bond and yielding a new radical. The net result is a radical addition reaction.



An example of an industrially useful radical reaction is the reaction of chlorine with methane to yield chloromethane, which is used to manufacture the solvents dichloromethane (CH_2Cl_2) and chloroform ($CHCl_3$). The process begins with irradiation of Cl_2 with ultraviolet light to break the relatively weak Cl–Cl bond of Cl_2 and produce chlorine radicals (·Cl).

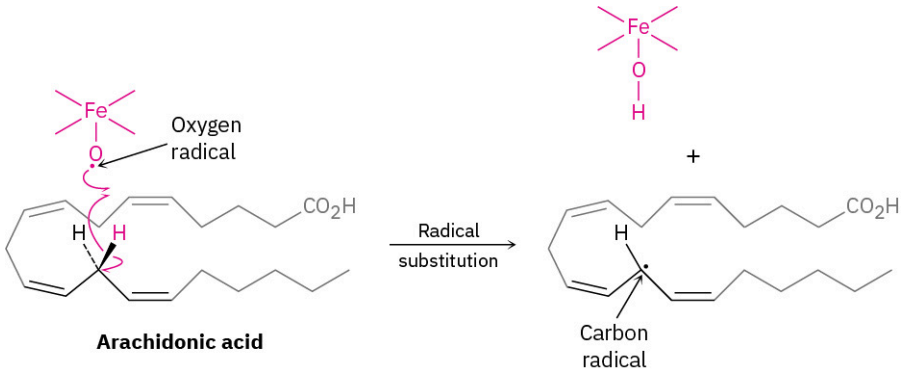


Chlorine radicals then react with methane by abstracting a hydrogen atom to give HCl and a methyl radical ($\cdot CH_3$) that reacts further with Cl_2 to give chloromethane plus a new chlorine radical that cycles back and repeats the first step. Thus, once the sequence has started, it becomes a self-sustaining cycle of repeating steps (**a**) and (**b**), making the overall process a chain reaction.



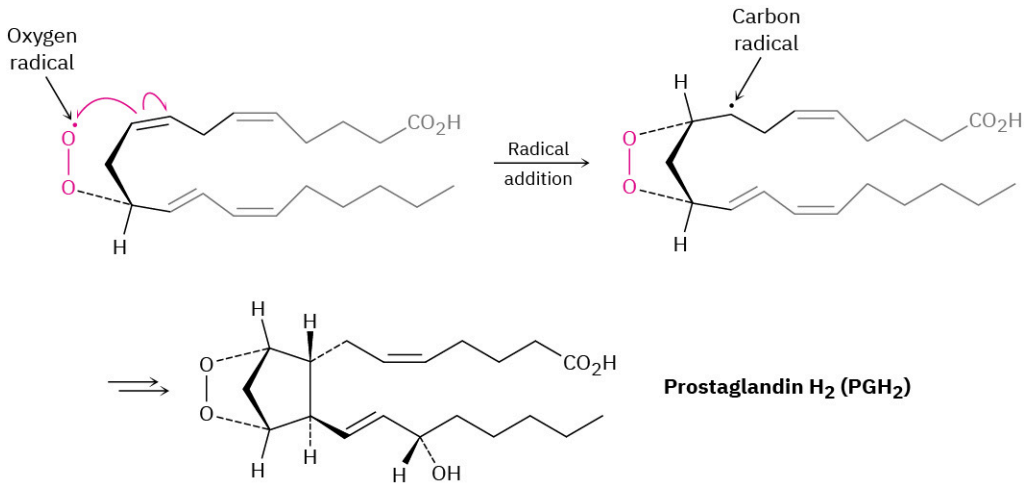
As a biological example of a radical reaction, look at the synthesis of *prostaglandins*, a large class of molecules found in virtually all body tissues and fluids. A number of pharmaceuticals are based on or derived from prostaglandins, including medicines that induce labor during childbirth, reduce intraocular pressure in glaucoma, control bronchial asthma, and help treat congenital heart defects.

Prostaglandin biosynthesis is initiated by the abstraction of a hydrogen atom from arachidonic acid by an iron–oxygen radical, thereby generating a carbon radical in a substitution reaction. Don't be intimidated by the size of the molecules; focus on the changes that occur in each step. (To help you do that, the unchanged part of the molecule is "ghosted," with only the reactive part clearly visible.)



Arachidonic acid

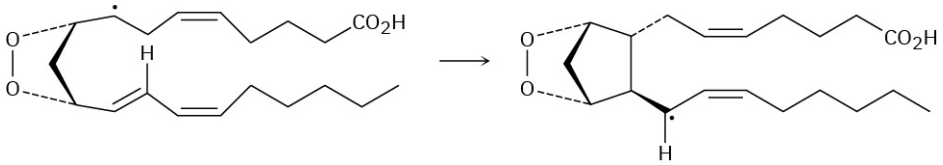
Following the initial abstraction of a hydrogen atom, the carbon radical then reacts with O_2 to give an oxygen radical, which reacts with a C=C bond within the same molecule in an addition reaction. Several further transformations ultimately yield prostaglandin H_2.



Prostaglandin H_2 (PGH_2)

PROBLEM 6-8 Radical chlorination of alkanes is not generally useful because mixtures of products often result when more than one kind of C–H bond is present in the substrate. Draw and name all monochloro substitution products $C_6H_{13}Cl$ you might obtain by reaction of 2-methylpentane with Cl_2.

PROBLEM 6-9 Using curved fishhook arrows, propose a mechanism for the formation of the cyclopentane ring of prostaglandin H_2.



6.7 Describing a Reaction: Equilibria, Rates, and Energy Changes

Every chemical reaction can go in either a forward or reverse direction. Reactants can go forward to products, and products can revert to reactants. As you may remember from your general chemistry course, the position of the resulting chemical equilibrium is expressed by an equation in which K_{eq}, the equilibrium constant, is equal to the product concentrations multiplied together, divided by the reactant concentrations multiplied together, with each concentration raised to the power of its coefficient in the balanced equation. For the generalized reaction

$$aA + bB \rightleftharpoons cC + dD$$

we have

$$K_{eq} = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

The value of the equilibrium constant tells which side of the reaction arrow is energetically favored. If K_{eq} is much larger than 1, then the product concentration term $[C]^c[D]^d$ is much larger than the reactant concentration term $[A]^a[B]^b$, and the reaction proceeds as written from left to right. If K_{eq} is near 1, appreciable amounts of both reactant and product are present at equilibrium. And if K_{eq} is much smaller than 1, the reaction does not take place as written but instead goes in the reverse direction, from right to left.

In the reaction of ethylene with HBr, for example, we can write the following equilibrium expression and determine experimentally that the equilibrium constant at room temperature is approximately 7.1×10^7:

$$H_2C{=}CH_2 + HBr \rightleftharpoons CH_3CH_2Br$$

$$K_{eq} = \frac{[CH_3CH_2Br]}{[H_2C{=}CH_2]\,[HBr]} = 7.1 \times 10^7$$

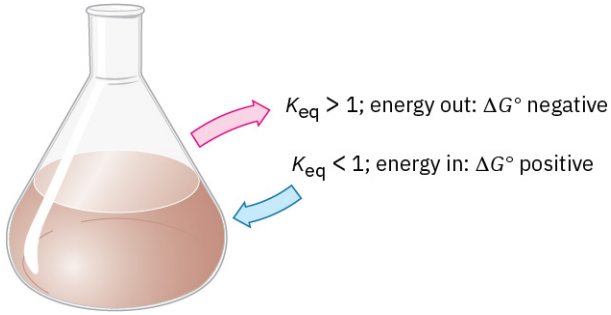
Because K_{eq} is relatively large, the reaction proceeds as written and more than 99.999 99% of the ethylene is converted into bromoethane. For practical purposes, an equilibrium constant greater than about 10^3 means that the amount of reactant left over will be barely detectable (less than 0.1%).

What determines the magnitude of the equilibrium constant? For a reaction to have a favorable equilibrium constant and proceed as written, the energy of the products must be lower than the energy of the reactants. In other words, energy must be released. This situation is analogous to that of a rock poised precariously in a high-energy position near the top of a hill. When it rolls downhill, the rock releases energy until it reaches a more stable, low-energy position at the bottom.

The energy change that occurs during a chemical reaction is called the **Gibbs free-energy change (ΔG)**, which is equal to the free energy of the products minus the free energy of the reactants: $\Delta G = G_{products} - G_{reactants}$. For a favorable reaction, ΔG has a negative value, meaning that energy is lost by the chemical system and released *to* the surroundings, usually as heat. Such reactions are said to be **exergonic**. For an unfavorable reaction, ΔG has a positive value, meaning that energy is absorbed by the chemical system *from* the surroundings. Such reactions are said to be **endergonic**.

You might also recall from general chemistry that the *standard* free-energy change for a reaction is denoted as $\Delta G°$, where the superscript ° means that the reaction is carried out under standard conditions, with pure substances in their most stable form at 1 atm pressure and a specified temperature, usually 298 K. For biological reactions, the standard free-energy change is denoted as $\Delta G°'$ and refers to a reaction carried out at

pH = 7.0 with solute concentrations of 1.0 M.



Because the equilibrium constant, K_{eq}, and the standard free-energy change, $\Delta G°$, both measure whether a reaction is favorable, they are mathematically related by the equation

$$\Delta G^\circ = -RT \ln K_{eq} \quad \text{or} \quad K_{eq} = e^{-\Delta G^\circ/RT}$$

where

$R = 8.314\ \text{J/(K} \cdot \text{mol)} = 1.987\ \text{cal/(K} \cdot \text{mol)}$
$T =$ Kelvin temperature
$e = 2.718$
$\ln K_{eq} =$ natural logarithm of K_{eq}

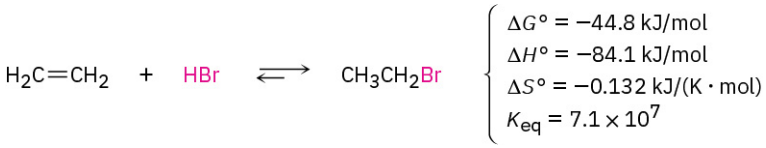
For example, the reaction of ethylene with HBr has $K_{eq} = 7.1 \times 10^7$, so $\Delta G° = -44.8$ kJ/mol (−10.7 kcal/mol) at 298 K:

$$K_{eq} = 7.1 \times 10^7 \quad \text{and} \quad \ln K_{eq} = 18.08$$
$$\Delta G^\circ = -RT \ln K_{eq} = -[8.314\ \text{J/(K} \cdot \text{mol)}](298\ \text{K})(18.08)$$
$$= -44{,}800\ \text{J/mol} = -44.8\ \text{kJ/mol}$$

The free-energy change ΔG is made up of two terms, an *enthalpy* term, ΔH, and a temperature-dependent *entropy* term, $T\Delta S$. Of the two terms, the enthalpy term is often larger and more dominant.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

For the reaction of ethylene with HBr at room temperature (298 K), the approximate values are



The **enthalpy change (ΔH)**, also called the **heat of reaction**, is a measure of the change in total bonding energy during a reaction. If ΔH is negative, as in the reaction of HBr with ethylene, the products have less energy than the reactants. Thus, the products are more stable and have stronger bonds than the reactants, heat is released, and the reaction is said to be **exothermic**. If ΔH is positive, the products are less stable and have weaker bonds than the reactants, heat is absorbed, and the reaction is said to be **endothermic**. For example, if a reaction breaks reactant bonds with a total strength of 380 kJ/mol and forms product bonds with a total strength of 400 kJ/mol, then ΔH for the reaction is 380 kJ/mol - 400 kJ/mol = −20 kJ/mol and the reaction is exothermic.

The **entropy change (ΔS)** is a measure of the change in the amount of molecular randomness, or freedom of motion, that accompanies a reaction. For example, in an elimination reaction of the type

$$\mathbf{A} \longrightarrow \mathbf{B} + \mathbf{C}$$

there is more freedom of movement and molecular randomness in the products than in the reactant because one molecule has split into two. Thus, there is a net increase in entropy during the reaction and ΔS has a positive value.

On the other hand, for an addition reaction of the type

$$\mathbf{A + B \longrightarrow C}$$

the opposite is true. Because such reactions restrict the freedom of movement of two molecules by joining them together, the product has less randomness than the reactants and ΔS has a negative value. The reaction of ethylene and HBr to yield bromoethane, which has $\Delta S°$ = −0.132 kJ/(K · mol), is an example. TABLE 6.2 describes the thermodynamic terms more fully.

TABLE 6.2 **Explanation of Thermodynamic Quantities: $\Delta G° = \Delta H° - T\Delta S°$**

Term	Name	Explanation
$\Delta G°$	**Gibbs free-energy change**	The energy difference between reactants and products. When $\Delta G°$ is negative, the reaction is **exergonic**, has a favorable equilibrium constant, and can occur spontaneously. When $\Delta G°$ is positive, the reaction is **endergonic**, has an unfavorable equilibrium constant, and cannot occur spontaneously.
$\Delta H°$	**Enthalpy change**	The heat of reaction, or difference in strength between the bonds broken in a reaction and the bonds formed. When $\Delta H°$ is negative, the reaction releases heat and is **exothermic**. When $\Delta H°$ is positive, the reaction absorbs heat and is **endothermic**.
$\Delta S°$	**Entropy change**	The change in molecular randomness during a reaction. When $\Delta S°$ is negative, randomness decreases. When $\Delta S°$ is positive, randomness increases.

Knowing the value of K_{eq} for a reaction is useful, but it's important to realize its limitations. An equilibrium constant tells only the position of the equilibrium, or how much product is theoretically possible. It doesn't tell the rate of reaction, or how fast the equilibrium is established. Some reactions are extremely slow even though they have favorable equilibrium constants. Gasoline is stable at room temperature, for instance, because the rate of its reaction with oxygen is slow at 298 K. Only at higher temperatures, such as contact with a lighted match, does gasoline react rapidly with oxygen and undergo complete conversion to the equilibrium products water and carbon dioxide. Rates (how fast a reaction occurs) and equilibria (how much a reaction occurs) are entirely different.

Rate → Is the reaction fast or slow?
Equilibrium → In what direction does the reaction proceed?

PROBLEM 6-10 Which reaction is more energetically favored, one with $\Delta G°$ = −44 kJ/mol or one with $\Delta G°$ = +44 kJ/mol?

PROBLEM 6-11 Which reaction is more exergonic, one with K_{eq} = 1000 or one with K_{eq} = 0.001?

6.8 Describing a Reaction: Bond Dissociation Energies

We've just seen that heat is released (negative ΔH) when a bond is formed because the products are more stable and have stronger bonds than the reactants. Conversely, heat is absorbed (positive ΔH) when a bond is broken because the products are less stable and have weaker bonds than the reactants. The amount of energy needed to break a given bond to produce two radical fragments when the molecule is in the gas phase at 25 °C is a quantity called the **bond strength**, or **bond dissociation energy (*D*)**.

$$A{:}B \xrightarrow[\text{energy}]{\text{Bond dissociation}} A\cdot + \cdot B$$

Each specific bond has its own characteristic strength, and extensive tables of such data are available. For example, a C–H bond in methane has a bond dissociation energy D = 439.3 kJ/mol (105.0 kcal/mol), meaning that 439.3 kJ/mol must be added to break a C–H bond of methane to give the two radical fragments $\cdot CH_3$ and ·H. Conversely, 439.3 kJ/mol of energy is released when a methyl radical and a hydrogen atom combine to form methane. TABLE 6.3 lists some other bond strengths.

TABLE 6.3 Some Bond Dissociation Energies, *D*

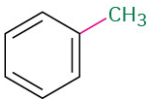
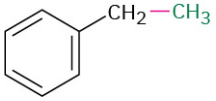
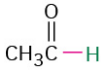
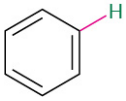
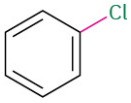
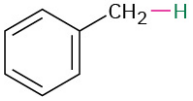
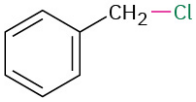
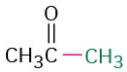
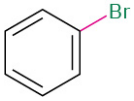
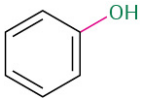
Bond	*D* (kJ/mol)	Bond	*D* (kJ/mol)	Bond	*D* (kJ/mol)
H—H	436	$(CH_3)_2CH$—H	410	C_2H_5—CH_3	370
H—F	570	$(CH_3)_2CH$—Cl	354	$(CH_3)_2CH$—CH_3	369
H—Cl	431	$(CH_3)_2CH$—Br	299	$(CH_3)_3C$—CH_3	363
H—Br	366	$(CH_3)_3C$—H	400	H_2C=CH—CH_3	426
H—I	298	$(CH_3)_3C$—Cl	352	H_2C=$CHCH_2$—CH_3	318
Cl—Cl	242	$(CH_3)_3C$—Br	293	H_2C=CH_2	728
Br—Br	194	$(CH_3)_3C$—I	227		427
I—I	152	H_2C=CH—H	464		325
CH_3—H	439	H_2C=CH—Cl	396		374
CH_3—Cl	350	H_2C=$CHCH_2$—H	369	HO—H	497
CH_3—Br	294	H_2C=$CHCH_2$—Cl	298	HO—OH	211
CH_3—I	239		472	CH_3O—H	440
CH_3—OH	385		400	CH_3S—H	366
CH_3—NH_2	386		375	C_2H_5O—H	441
C_2H_5—H	421		300		352
C_2H_5—Cl	352		336	CH_3CH_2O—CH_3	355

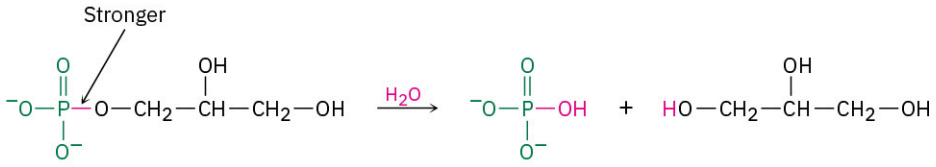
TABLE 6.3 Some Bond Dissociation Energies, *D*

Bond	*D* (kJ/mol)	Bond	*D* (kJ/mol)	Bond	*D* (kJ/mol)
C_2H_5—Br	293		464	NH_2—H	450
C_2H_5—I	233	HC≡C—H	558	H—CN	528
C_2H_5—OH	391	CH_3—CH_3	377		

Think again about the connection between bond strengths and chemical reactivity. In an exothermic reaction, more heat is released than is absorbed. But because making bonds in the products releases heat and breaking bonds in the reactants absorbs heat, the bonds in the products must be stronger than the bonds in the reactants. In other words, exothermic reactions are favored by products with strong bonds and by reactants with weak, easily broken bonds.

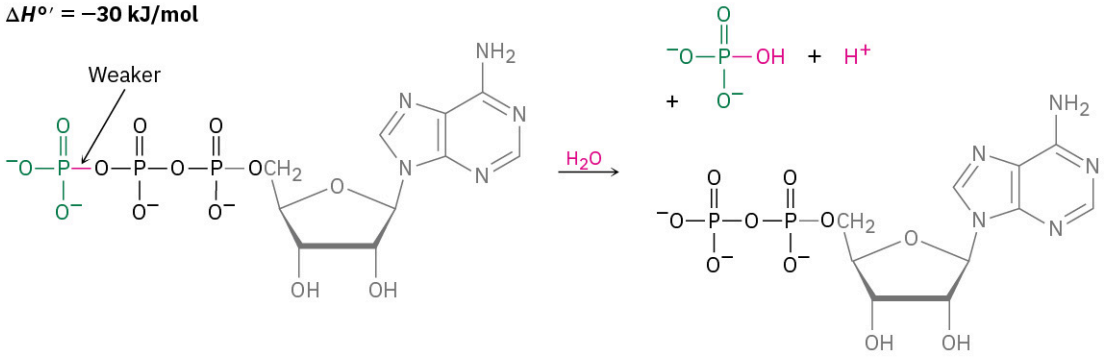
Sometimes, particularly in biochemistry, reactive substances that undergo highly exothermic reactions, such as ATP (adenosine triphosphate), are referred to as "energy-rich" or "high-energy" compounds. Such a label doesn't mean that ATP is special or different from other compounds, it only means that ATP has relatively weak bonds that require a relatively small amount of heat to break, thus leading to a larger release of heat when a strong new bond forms in a reaction. When a typical organic phosphate such as glycerol 3-phosphate reacts with water, for instance, only 9 kJ/mol of heat is released ($\Delta H^{o\prime} = -9$ kJ/mol), but when ATP reacts with water, 30 kJ/mol of heat is released ($\Delta H^{o\prime} = -30$ kJ/mol). The difference between the two reactions is due to the fact that the bond broken in ATP is substantially weaker than the bond broken in glycerol 3-phosphate. We'll see the metabolic importance of this reaction in later chapters.

$\Delta H^{o\prime} = -9$ kJ/mol



Glycerol 3-phosphate **Glycerol**

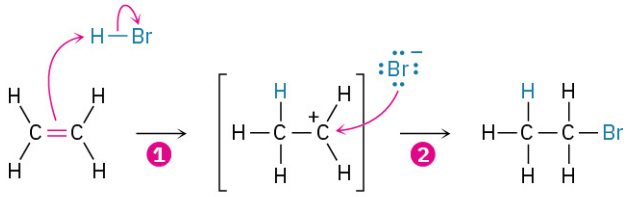
$\Delta H^{o\prime} = -30$ kJ/mol



Adenosine triphosphate (ATP) **Adenosine diphosphate (ADP)**

6.9 Describing a Reaction: Energy Diagrams and Transition States

For a reaction to take place, reactant molecules must collide and reorganization of atoms and bonds must occur. Let's again look at the addition reaction of HBr and ethylene.



As the reaction proceeds, ethylene and HBr approach each other, the ethylene π bond and the H–Br bond break, a new C–H bond forms in step **1** and a new C–Br bond forms in step **2**.

To depict graphically the energy changes that occur during a reaction, chemists use energy diagrams, such as that in **FIGURE 6.5**. The vertical axis of the diagram represents the total energy of all reactants, and the horizontal axis, called the **reaction coordinate**, represents the progress of the reaction from beginning to end. Let's see how the addition of HBr to ethylene can be described in an energy diagram.

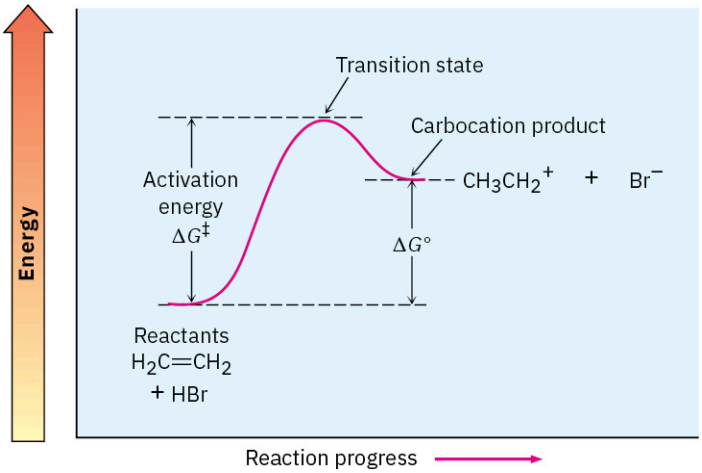


FIGURE 6.5 An energy diagram for the first step in the reaction of ethylene with HBr. The energy difference between reactants and the transition state, $\Delta G^{\ddagger}$, defines the reaction rate. The energy difference between reactants and carbocation product, $\Delta G°$, defines the position of the equilibrium.

At the beginning of the reaction, ethylene and HBr have the total amount of energy indicated by the reactant level on the left side of the diagram in **FIGURE 6.5**. As the two reactants collide and reaction commences, their electron clouds repel each other, causing the energy level to rise. If the collision has occurred with enough force and proper orientation, however, the reactants continue to approach each other despite the rising repulsion until the new C–H bond starts to form. At some point, a structure of maximum energy is reached, a structure called the **transition state**.

The transition state represents the highest-energy structure involved in this step of the reaction. It is unstable and can't be isolated, but we can imagine it to be an activated complex of the two reactants in which both the C=C π bond and H–Br bond are partially broken and the new C–H bond is partially formed (**FIGURE 6.6**).

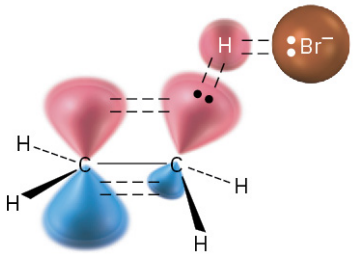


FIGURE 6.6 A hypothetical transition-state structure for the first step of the reaction of ethylene with HBr. The C=C π bond and H–Br bond are just beginning to break, and the C–H bond is just beginning to form.

The energy difference between reactants and the transition state is called the **activation energy, $\Delta G^{\ddagger}$**, and determines how rapidly the reaction occurs at a given temperature. (The double-dagger superscript, ‡, always

refers to the transition state.) A high activation energy results in a slow reaction because few collisions occur with enough energy for the reactants to reach the transition state. A low activation energy results in a rapid reaction because almost all collisions occur with enough energy for the reactants to reach the transition state.

As an analogy, you might think of reactants that need enough energy to climb the activation barrier to the transition state as hikers who need enough energy to climb to the top of a mountain pass. If the pass is a high one, the hikers need a lot of energy and surmount the barrier with difficulty. If the pass is low, the hikers need less energy and reach the top easily.

As a rough generalization, many organic reactions have activation energies in the range 40 to 150 kJ/mol (10–35 kcal/mol). The reaction of ethylene with HBr, for example, has an activation energy of approximately 140 kJ/mol (34 kcal/mol). Reactions with activation energies less than 80 kJ/mol take place at or below room temperature, while reactions with higher activation energies normally require a higher temperature to give the reactants enough energy to climb the activation barrier.

Once the transition state is reached, the reaction can either continue on to give the carbocation product or revert back to reactants. When reversion to reactants occurs, the transition-state structure comes apart and an amount of free energy corresponding to $-\Delta G^{\ddagger}$ is released. When the reaction continues on to give the carbocation, the new C–H bond forms fully and an amount of energy is released corresponding to the difference between the transition state and carbocation product. The net energy change for the step, $\Delta G°$, is represented in the diagram as the difference in level between reactant and product. Since the carbocation is higher in energy than the starting alkene, the step is endergonic, has a positive value of $\Delta G°$, and absorbs energy.

Not all energy diagrams are like that shown for the reaction of ethylene and HBr. Each reaction has its own energy profile. Some reactions are fast (small $\Delta G^{\ddagger}$) and some are slow (large $\Delta G^{\ddagger}$); some have a negative $\Delta G°$, and some have a positive $\Delta G°$. FIGURE 6.7 illustrates some different possibilities.

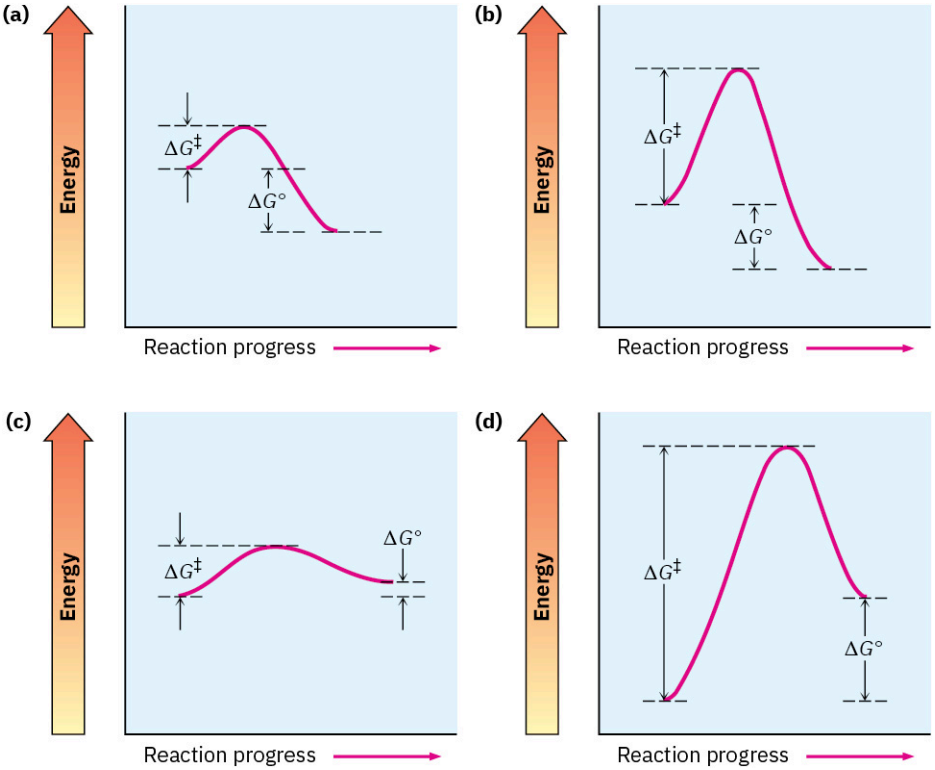
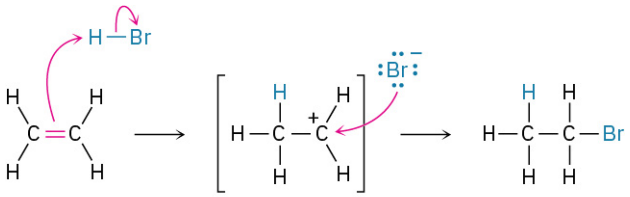


FIGURE 6.7 **Some hypothetical energy diagrams: (a)** a fast exergonic reaction (low $\Delta G^{\ddagger}$, negative $\Delta G°$); **(b)** a slow exergonic reaction (high $\Delta G^{\ddagger}$, negative $\Delta G°$); **(c)** a fast endergonic reaction (small $\Delta G^{\ddagger}$, small positive $\Delta G°$); **(d)** a slow endergonic reaction (high $\Delta G^{\ddagger}$, positive $\Delta G°$).

PROBLEM 6-12 Which reaction is faster, one with $\Delta G^{\ddagger} = +45$ kJ/mol or one with $\Delta G^{\ddagger} = +70$ kJ/mol?

6.10 Describing a Reaction: Intermediates

How can we describe the carbocation formed in the first step of the reaction of ethylene with HBr? The carbocation is clearly different from the reactants, yet it isn't a transition state and it isn't a final product.



Reaction intermediate

We call the carbocation, which exists only transiently during the course of the multistep reaction, a **reaction intermediate**. As soon as the intermediate is formed in the first step by reaction of ethylene with H^+, it reacts further with Br^- in a second step to give the final product, bromoethane. This second step has its own activation energy ($\Delta G^{\ddagger}$), its own transition state, and its own energy change ($\Delta G°$). We can picture the second transition state as an activated complex between the electrophilic carbocation intermediate and the nucleophilic bromide anion, in which Br^- donates a pair of electrons to the positively charged carbon atom as the new C–Br bond just starts to form.

A complete energy diagram for the overall reaction of ethylene with HBr is shown in **FIGURE 6.8**. In essence, we draw a diagram for each of the individual steps and then join them so that the carbocation product of step 1 is the reactant for step 2. As indicated in **FIGURE 6.8**, the reaction intermediate lies at an energy minimum between steps. Because the energy level of the intermediate is higher than the level of either the reactant that formed it or the product it yields, the intermediate can't normally be isolated. It is, however, more stable than its two neighboring transition states.

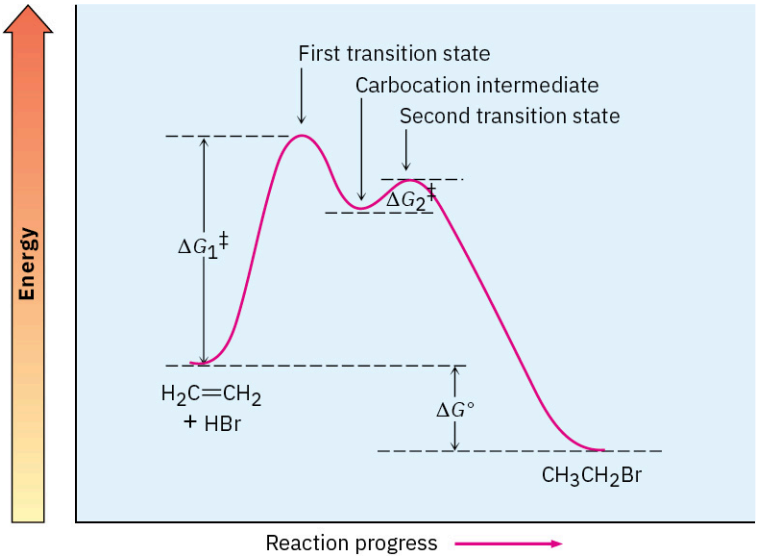


FIGURE 6.8 An energy diagram for the reaction of ethylene with HBr. Two separate steps are involved, each with its own activation energy ($\Delta G^{\ddagger}$) and free-energy change ($\Delta G°$). The overall $\Delta G^{\ddagger}$ for the complete reaction is the energy difference between reactants and the highest transition state (which corresponds to $\Delta G1^{\ddagger}$ in this case), and the overall $\Delta G°$ for the reaction is the energy difference between reactants and final products.

Each step in a multistep process can always be considered separately. Each step has its own $\Delta G^{\ddagger}$ and its own $\Delta G°$. The overall activation energy that controls the rate of the reaction, however, is the energy difference between initial reactants and the highest transition state, regardless of which step it occurs in. The overall $\Delta G°$ of the reaction is the energy difference between reactants and final products.

The biological reactions that take place in living organisms have the same energy requirements as reactions that take place in the laboratory and can be described in similar ways. They are, however, constrained by the fact that they must have low enough activation energies to occur at moderate temperatures, and they must release

energy in relatively small amounts to avoid overheating the organism. These constraints can be met through the use of large, structurally complex, enzyme catalysts that alter the mechanism of a reaction to a pathway that can proceed through a series of small steps rather than one or two large steps. Thus, an energy diagram for a biological reaction might look like that in FIGURE 6.9.

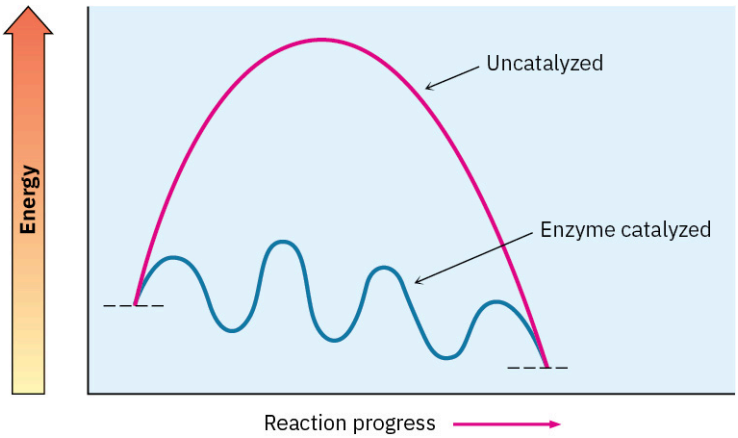


FIGURE 6.9 Energy diagrams for a typical, enzyme-catalyzed biological reaction and an uncatalyzed laboratory reaction. The biological reaction involves many steps, each of which has a relatively small activation energy and small energy change. The end result is the same, however.

WORKED EXAMPLE 6.3

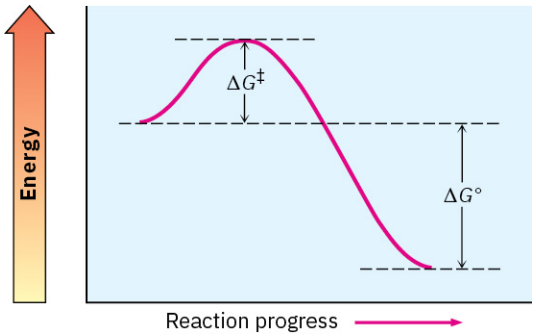
Drawing a Reaction Energy Diagram

Sketch an energy diagram for a one-step reaction that is fast and highly exergonic.

Strategy

A fast reaction has a small $\Delta G^{\ddagger}$, and a highly exergonic reaction has a large negative $\Delta G°$.

Solution



WORKED EXAMPLE 6.4

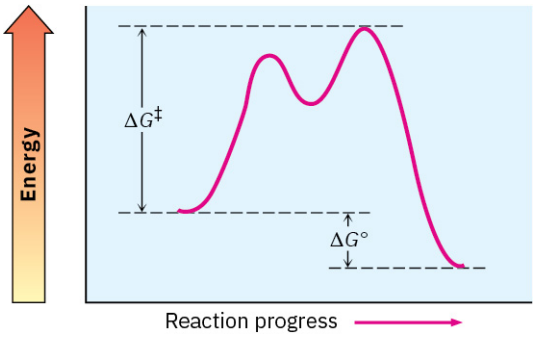
Drawing a Reaction Energy Diagram

Sketch an energy diagram for a two-step exergonic reaction whose second step has a higher-energy transition state than its first step. Show $\Delta G^{\ddagger}$ and $\Delta G°$ for the overall reaction.

Strategy

A two-step reaction has two transition states and an intermediate between them. The $\Delta G^{\ddagger}$ for the overall reaction is the energy change between reactants and the highest-energy transition state—the second one in this case. An exergonic reaction has a negative overall $\Delta G°$.

Solution

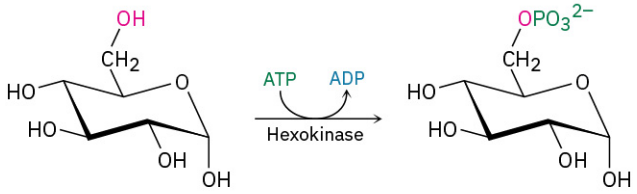


PROBLEM 6-13 Sketch an energy diagram for a two-step reaction in which both steps are exergonic and in which the second step has a higher-energy transition state than the first. Label the parts of the diagram corresponding to reactant, product, intermediate, overall $\Delta G^{\ddagger}$, and overall $\Delta G°$.

6.11 A Comparison Between Biological Reactions and Laboratory Reactions

Beginning in the next chapter, we'll be seeing a lot of reactions, some that are important in laboratory chemistry yet don't occur in nature and others that have counterparts in biological pathways. In comparing laboratory reactions with biological reactions, several differences are apparent. For one, laboratory reactions are usually carried out in an organic solvent such as diethyl ether or dichloromethane to dissolve the reactants and bring them into contact, whereas biological reactions occur in the aqueous medium within cells. For another, laboratory reactions often take place over a wide range of temperatures without catalysts, while biological reactions take place at the temperature of the organism and are catalyzed by enzymes.

We'll look at **enzymes** in more detail in **Section 26.10**, but you may already be aware that an enzyme is a large, globular, protein molecule that contains in its structure a protected pocket called its **active site**. The active site is lined by acidic or basic groups as needed for catalysis and has precisely the right shape to bind and hold a substrate molecule in the orientation necessary for reaction. FIGURE 6.10 shows a molecular model of hexokinase, along with an X-ray crystal structure of the glucose substrate and adenosine diphosphate (ADP) bound in the active site. Hexokinase is an enzyme that catalyzes the initial step of glucose metabolism—the transfer of a phosphate group from ATP to glucose, giving glucose 6-phosphate and ADP. The structures of ATP and ADP were shown at the end of Section 6.8.



Glucose **Glucose 6-phosphate**

Note how the hexokinase-catalyzed phosphorylation reaction of glucose is written. It's common when writing biological equations to show only the structures of the primary reactant and product, while abbreviating the structures of various biological "reagents" and by-products such as ATP and ADP. A curved arrow intersecting the straight reaction arrow indicates that ATP is also a reactant and ADP also a product.

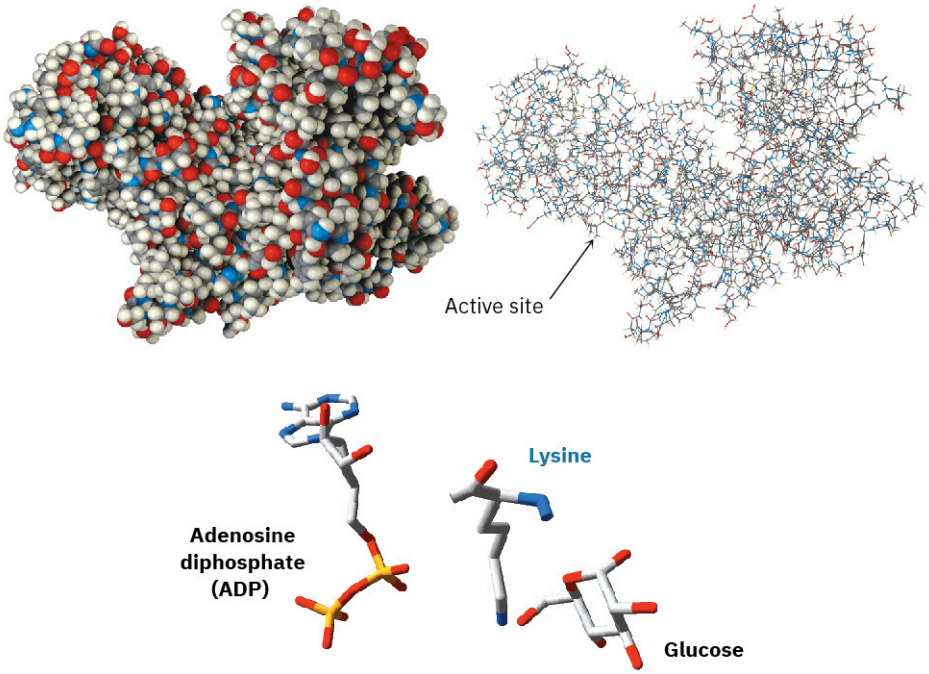
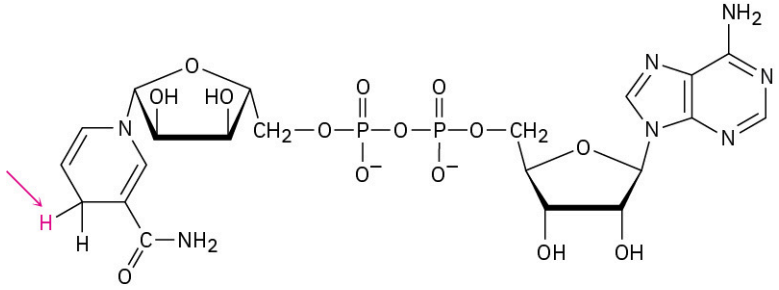


FIGURE 6.10 **Models of hexokinase in space-filling and wire-frame formats, showing the cleft that contains the active site where substrate binding and reaction catalysis occur.** At the bottom is an X-ray crystal structure of the enzyme active site, showing the positions of both glucose and ADP as well as a lysine amino acid that acts as a base to deprotonate glucose.

Yet a third difference between laboratory and biological reactions is that laboratory reactions are often done using relatively small, simple reagents such as Br_2, HCl, $NaBH_4$, CrO_3, and so forth, while biological reactions usually involve relatively complex "reagents" called *coenzymes*. In the hexokinase-catalyzed phosphorylation of glucose just shown, ATP is the coenzyme. As another example, compare the H_2 molecule, a laboratory reagent that adds to a carbon–carbon double bond to yield an alkane, with the reduced nicotinamide adenine dinucleotide (NADH) molecule, a coenzyme that effects an analogous addition of hydrogen to a double bond in many biological pathways. Of all the atoms in the coenzyme, only the one hydrogen atom shown in red is transferred to the double-bond substrate.



Reduced nicotinamide adenine dinucleotide, NADH (a coenzyme)

Don't be intimidated by the size of the ATP or NADH molecule; most of the structure is there to provide an overall shape for binding to the enzyme and to provide appropriate solubility behavior. When looking at biological molecules, focus on the small part of the molecule where the chemical change takes place.

One final difference between laboratory and biological reactions is in their specificity. A catalyst might be used in the laboratory to catalyze the reaction of thousands of different substances, but an enzyme, because it can only bind a specific substrate molecule having a specific shape, will usually catalyze only a specific reaction. It's this exquisite specificity that makes biological chemistry so remarkable and that makes life possible. TABLE 6.4 summarizes some of the differences between laboratory and biological reactions.

TABLE 6.4 A Comparison of Typical Laboratory and Biological Reactions A Comparison of Typical Laboratory and Biological Reactions

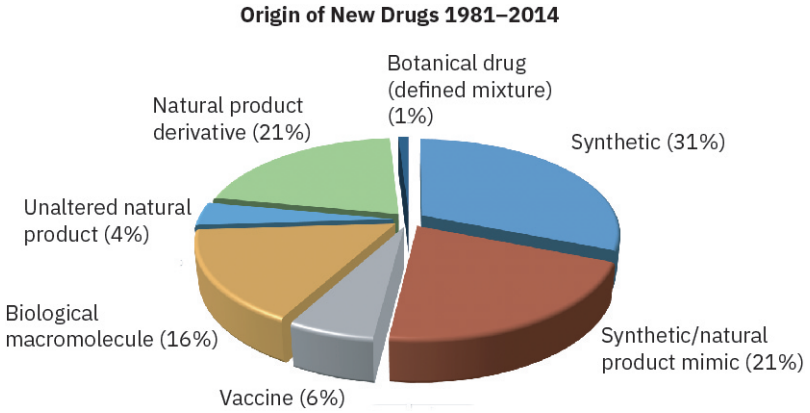
	Laboratory reaction	Biological reaction
Solvent	Organic liquid, such as ether	Aqueous environment in cells
Temperature	Wide range; −80 to 150 °C	Temperature of organism
Catalyst	Either none, or very simple	Large, complex enzymes needed
Reagent size	Usually small and simple	Relatively complex coenzymes
Specificity	Little specificity for substrate	Very high specificity for substrate

CHEMISTRY MATTERS

Where Do Drugs Come From?

It has been estimated that major pharmaceutical companies in the United States spent some $200 billion on drug research and development in 2020, while government agencies and private foundations spent another $28 billion. What does this money buy? From 1983 to 2022, the money resulted in a total of 1237 new molecular entities (NMEs)—new biologically active chemical substances approved for sale as drugs by the U.S. Food and Drug Administration (FDA).

Where do the new drugs come from? According to a study carried out several years ago at the U.S. National Cancer Institute, only about 33% of new drugs are entirely synthetic and completely unrelated to any naturally occurring substance. The remaining 67% take their lead, to a greater or lesser extent, from nature. Vaccines and genetically engineered proteins of biological origin account for 15% of NMEs, but most new drugs come from *natural products*, a catchall term generally taken to mean small molecules found in bacteria, plants, algae, and other living organisms. Unmodified natural products isolated directly from the producing organism account for 24% of NMEs, while natural products that have been chemically modified in the laboratory account for the remaining 28%.



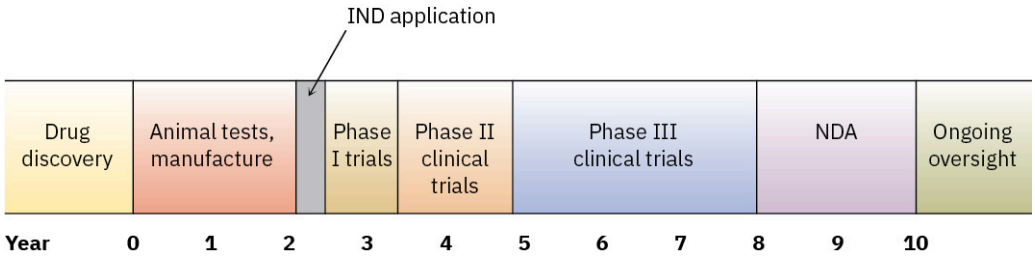
Many years of work go into screening many thousands of substances to identify a single compound that might ultimately gain approval as an NME. But after that single compound has been identified, the work has just begun because it takes an average of 9 to 10 years for a drug to make it through the approval process. First, the safety of the drug in animals must be demonstrated and an economical method of manufacture must be devised. With these preliminaries out of the way, an Investigational New Drug (IND) application is submitted to the FDA for permission to begin testing in humans.



FIGURE 6.11 Introduced in June, 2006, Gardasil is the first vaccine ever approved for the prevention of cancer. Where do new drugs like this come from? (credit: modification of work "COVIran Barekat vaccine production" by Sadegh Nikgostar/Wikimedia Commons, CC BY 4.0)

Human testing takes, or should take, 5 to 7 years and is divided into three phases. Phase I clinical trials are

carried out on a small group of healthy volunteers to establish safety and look for side effects. Several months to a year are needed, and only about 70% of drugs pass at this point. Phase II clinical trials next test the drug for 1 to 2 years in several hundred patients with the target disease or condition, looking both for safety and efficacy, and only about 33% of the original group pass. Finally, phase III trials are undertaken on a large sample of patients to document definitively the drug's safety, dosage, and efficacy. If the drug is one of the 25% of the original group that make it to the end of phase III, all the data are then gathered into a New Drug Application (NDA) and sent to the FDA for review and approval, which can take another 2 years. Ten years have elapsed and at least $500 million has been spent, with only a 20% success rate for the drugs that began testing. Finally, though, the drug will begin to appear in medicine cabinets. The following timeline shows the process.



Key Terms

- **activation energy ($\Delta G^{\ddagger}$)**
- **active site**
- **addition reaction**
- **bond dissociation energy (D)**
- **carbocation**
- **electrophile**
- **elimination reaction**
- **endergonic**
- **endothermic**
- **enthalpy change (ΔH)**
- **entropy change (ΔS)**
- **enzyme**
- **exergonic**
- **exothermic**
- **Gibbs free-energy change (ΔG)**
- **heat of reaction**
- **nucleophile**
- **polar reaction**
- **polarizability**
- **radical**
- **radical reaction**
- **reaction coordinate**
- **reaction intermediate**
- **reaction mechanism**
- **rearrangement reaction**
- **substitution reaction**
- **transition state**

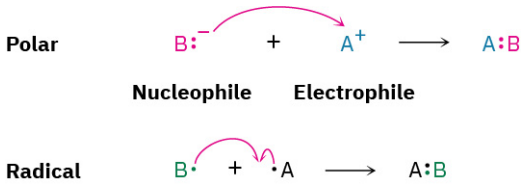
Summary

All chemical reactions, whether in the laboratory or in living organisms, follow the same chemical rules. To understand both organic and biological chemistry, it's necessary to know not just *what* occurs but also *why* and *how* chemical reactions take place. In this chapter, we've taken a brief look at the fundamental kinds of organic reactions, we've seen why reactions occur, and we've seen how reactions can be described.

There are four common kinds of reactions: **addition reactions** take place when two reactants add together to give a single product; **elimination reactions** take place when one reactant splits apart to give two products; **substitution reactions** take place when two reactants exchange parts to give two new products; and **rearrangement reactions** take place when one reactant undergoes a reorganization of bonds and atoms to give an isomeric product.

A full description of how a reaction occurs is called its **mechanism**. There are two general kinds of mechanisms by which most reactions take place: **radical** mechanisms and **polar** mechanisms. Polar reactions, the more common type, occur because of an attractive interaction between a **nucleophilic** (electron-rich) site in one molecule and an **electrophilic** (electron-poor) site in another molecule. A bond is formed in a polar reaction when the nucleophile donates an electron pair to the electrophile. This transfer of electrons is indicated by a curved arrow showing the direction of electron travel from the nucleophile to the electrophile. Radical reactions involve species that have an odd number of electrons. A bond is formed when each reactant donates one

electron.



The energy changes that take place during reactions can be described by considering both rates (how fast the reactions occur) and equilibria (how much the reactions occur). The position of a chemical equilibrium is determined by the value of the **free-energy change (ΔG)** for the reaction, where $\Delta G = \Delta H - T\Delta S$. The **enthalpy** term (ΔH) corresponds to the net change in strength of chemical bonds broken and formed during the reaction; the **entropy** term (ΔS) corresponds to the change in the amount of molecular randomness during the reaction. Reactions that have negative values of ΔG release energy, are said to be **exergonic**, and have favorable equilibria. Reactions that have positive values of ΔG absorb energy, are said to be **endergonic**, and have unfavorable equilibria.

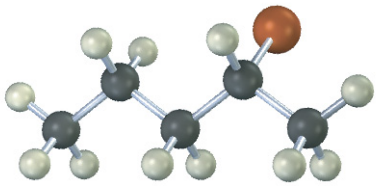
A reaction can be described pictorially using an energy diagram that follows the reaction course from reactants through transition state to product. The **transition state** is an activated complex occurring at the highest-energy point of a reaction. The amount of energy needed by reactants to reach this high point is the **activation energy, $\Delta G^{\ddagger}$**. The higher the activation energy, the slower the reaction.

Many reactions take place in more than one step and involve the formation of a **reaction intermediate**. An intermediate is a species that lies at an energy minimum between steps on the reaction curve and is formed briefly during the course of a reaction.

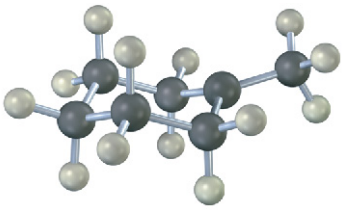
Additional Problems

Visualizing Chemistry

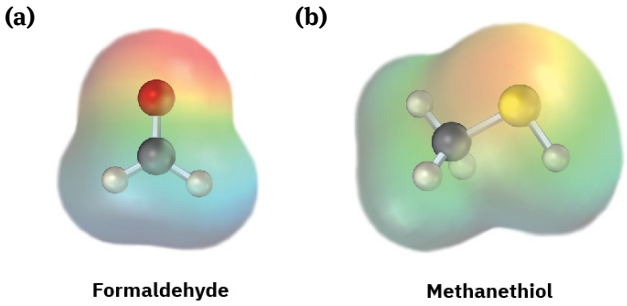
PROBLEM 6-14 The following alkyl halide can be prepared by addition of HBr to two different alkenes. Draw the structures of both (reddish-brown = Br).



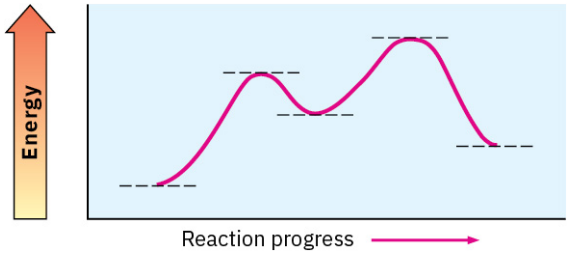
PROBLEM 6-15 The following structure represents the carbocation intermediate formed in the addition reaction of HBr to two different alkenes. Draw the structures of both.



PROBLEM 6-16 Electrostatic potential maps of **(a)** formaldehyde (CH_2O) and **(b)** methanethiol (CH_3SH) are shown. Is the formaldehyde carbon atom likely to be electrophilic or nucleophilic? What about the methanethiol sulfur atom? Explain.

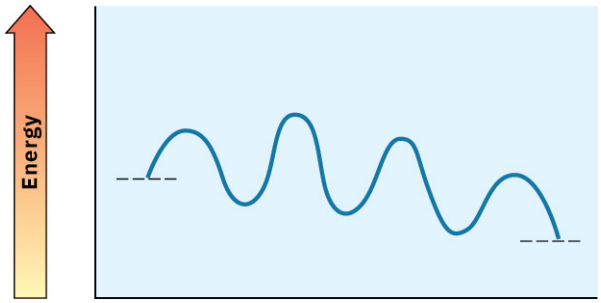


PROBLEM 6-17 Look at the following energy diagram:



(a) Is $\Delta G°$ for the reaction positive or negative? Label it on the diagram.
(b) How many steps are involved in the reaction?
(c) How many transition states are there? Label them on the diagram.

PROBLEM 6-18 Look at the following energy diagram for an enzyme-catalyzed reaction:



(a) How many steps are involved? **(b)** Which step is most exergonic?
(c) Which step is slowest?

Energy Diagrams and Reaction Mechanisms

PROBLEM 6-19 What is the difference between a transition state and an intermediate?

PROBLEM 6-20 Draw an energy diagram for a one-step reaction with $K_{eq} < 1$. Label the parts of the diagram corresponding to reactants, products, transition state, $\Delta G°$, and $\Delta G^{‡}$. Is $\Delta G°$ positive or negative?

PROBLEM 6-21 Draw an energy diagram for a two-step reaction with $K_{eq} > 1$. Label the overall $\Delta G°$, transition states, and intermediate. Is $\Delta G°$ positive or negative?

PROBLEM 6-22 Draw an energy diagram for a two-step exergonic reaction whose second step is faster than its first step.

PROBLEM 6-23 Draw an energy diagram for a reaction with $K_{eq} = 1$. What is the value of $\Delta G°$ in this reaction?

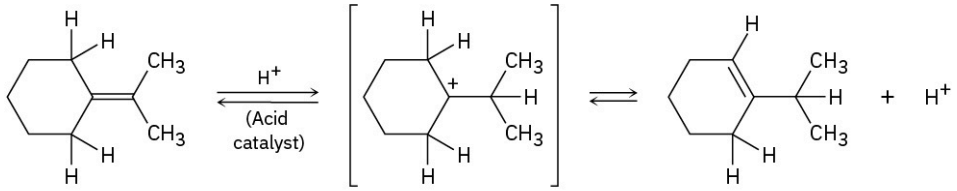
PROBLEM 6-24 The addition of water to ethylene to yield ethanol has the following thermodynamic parameters:

$H_2C{=}CH_2 + H_2O \rightleftharpoons CH_3CH_2OH$ $\left\{\begin{array}{l}\Delta H° = -44\ \text{kJ/mol}\\ \Delta S° = -0.12\ \text{kJ/(K · mol)}\\ K_{eq} = 24\end{array}\right.$

(a) Is the reaction exothermic or endothermic?

(b) Is the reaction favorable (spontaneous) or unfavorable (nonspontaneous) at room temperature (298 K)?

PROBLEM 6-25 When isopropylidenecyclohexane is treated with strong acid at room temperature, isomerization occurs by the mechanism shown below to yield 1-isopropylcyclohexene:



Isopropylidenecyclohexane **1-Isopropylcyclohexene**

At equilibrium, the product mixture contains about 30% isopropylidenecyclohexane and about 70% 1-isopropylcyclohexene.

(a) What is an approximate value of K_{eq} for the reaction?

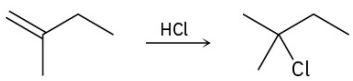
(b) Since the reaction occurs slowly at room temperature, what is its approximate $\Delta G^{\ddagger}$?

(c) Draw an energy diagram for the reaction.

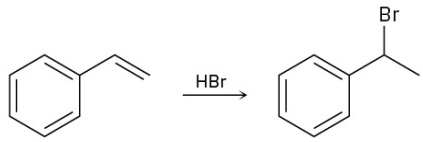
PROBLEM 6-26 Add curved arrows to the mechanism shown in Problem 6-25 to indicate the electron movement in each step.

PROBLEM 6-27 Draw the complete mechanism for each of the following polar reactions.

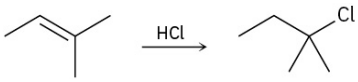
(a)



(b)

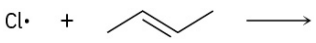


(c)

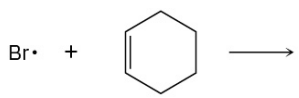


PROBLEM 6-28 Use curved arrows to show the flow of electrons, and draw the carbon radical that is formed when the halogen radicals below add to the corresponding alkenes.

(a)



(b)



(c)

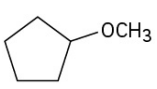


Polar Reactions

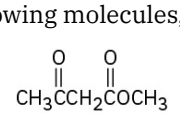
PROBLEM 6-29 Identify the functional groups in the following molecules, and show the polarity of each:

(a) $CH_3CH_2C{\equiv}N$

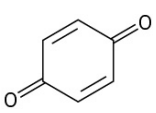
(b)



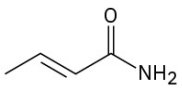
(c)



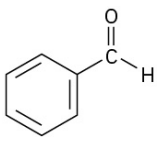
(d)



(e)



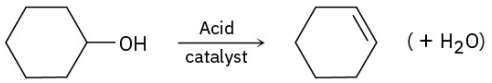
(f)



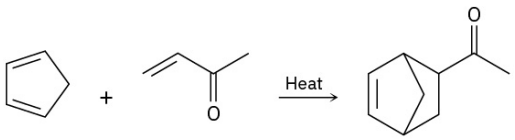
PROBLEM 6-30 Identify the following reactions as additions, eliminations, substitutions, or rearrangements:

(a) $CH_3CH_2Br + NaCN \longrightarrow CH_3CH_2CN\ (+ NaBr)$

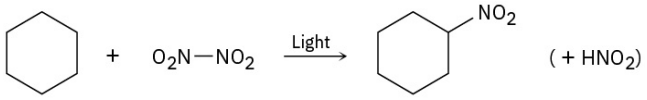
(b)



(c)

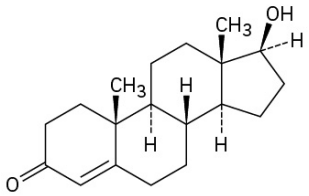


(d)



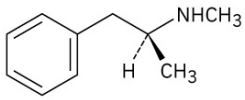
PROBLEM 6-31 Identify the likely electrophilic and nucleophilic sites in each of the following molecules:

(a)



Testosterone

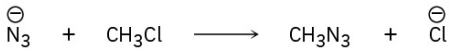
(b)



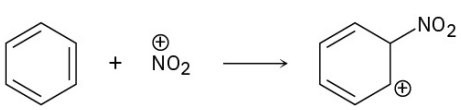
Methamphetamine

PROBLEM 6-32 Identify the electrophile and the nucleophile.

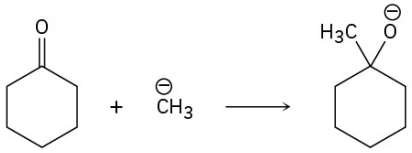
(a)



(b)

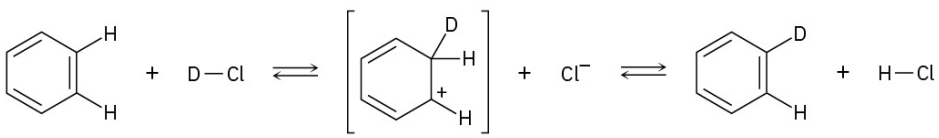


(c)

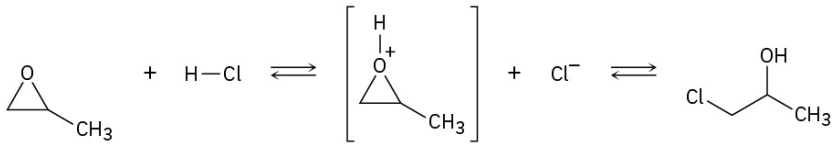


PROBLEM 6-33 Add curved arrows to the following polar reactions to indicate the flow of electrons in each:

(a)

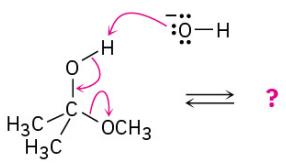


(b)

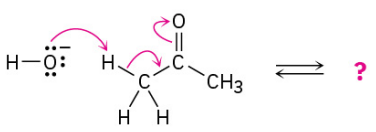


PROBLEM 6-34 Follow the flow of electrons indicated by the curved arrows in each of the following polar reactions, and predict the products that result:

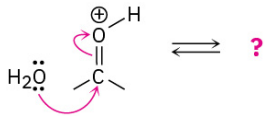
(a)



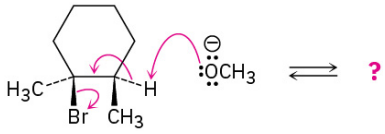
(b)



(c)



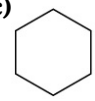
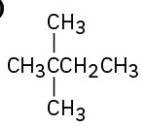
(d)



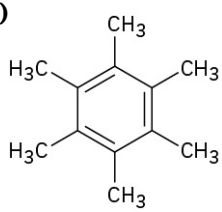
Radical Reactions

PROBLEM 6-35 Radical chlorination of pentane is a poor way to prepare 1-chloropentane, but radical chlorination of neopentane, $(CH_3)_4C$, is a good way to prepare neopentyl chloride, $(CH_3)_3CCH_2Cl$. Explain.

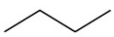
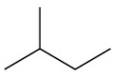
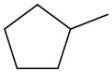
PROBLEM 6-36 Despite the limitations of radical chlorination of alkanes, the reaction is still useful for synthesizing certain halogenated compounds. For which of the following compounds does radical chlorination give a single monochloro product?

(a) CH_3CH_3 **(b)** $CH_3CH_2CH_3$ **(c)**  **(d)**  **(e)** $CH_3C{\equiv}CCH_3$

(f)

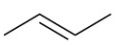
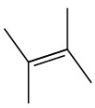


PROBLEM 6-37 Draw the different monochlorinated constitutional isomers you would obtain by the radical chlorination of the following compounds.

(a)  **(b)**  **(c)** 

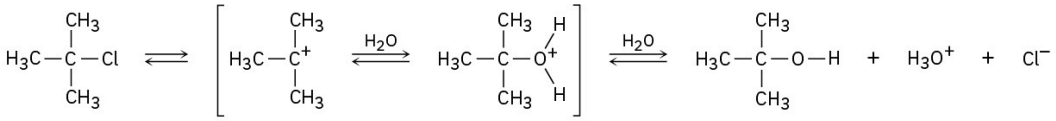
PROBLEM 6-38 Answer question 6-37 taking all stereoisomers into account.

PROBLEM 6-39 Show the structure of the carbocation that would result when each of the following alkenes reacts with an acid, H^+.

(a)  **(b)**  **(c)** 

General Problems

PROBLEM 6-40 2-Chloro-2-methylpropane reacts with water in three steps to yield 2-methyl-2-propanol. The first step is slower than the second, which in turn is much slower than the third. The reaction takes place slowly at room temperature, and the equilibrium constant is approximately 1.



2-Chloro-2-methylpropane **2-Methyl-2-propanol**

(a) Give approximate values for $\Delta G^{\ddagger}$ and ΔG° that are consistent with the above information.

(b) Draw an energy diagram for the reaction, labeling all points of interest and placing relative energy levels on the diagram consistent with the information given.

PROBLEM Add curved arrows to the mechanism shown in Problem 6-40 to indicate the electron movement in

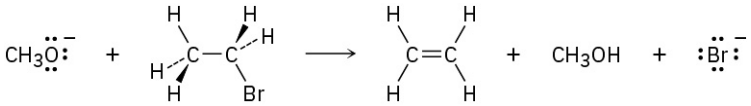
6-41 each step.

PROBLEM 6-42 The reaction of hydroxide ion with chloromethane to yield methanol and chloride ion is an example of a general reaction type called a nucleophilic substitution reaction:

$$HO^- + CH_3Cl \rightleftarrows CH_3OH + Cl^-$$

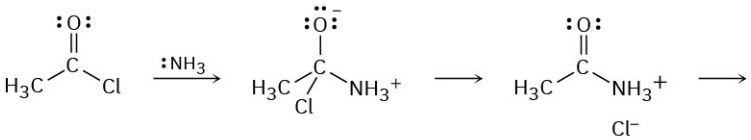
The value of $\Delta H°$ for the reaction is −75 kJ/mol, and the value of $\Delta S°$ is +54 J/(K·mol). What is the value of $\Delta G°$ (in kJ/mol) at 298 K? Is the reaction exothermic or endothermic? Is it exergonic or endergonic?

PROBLEM 6-43 Methoxide ion (CH_3O^-) reacts with bromoethane in a single step according to the following equation:

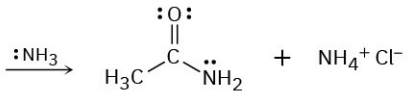


Identify the bonds broken and formed, and draw curved arrows to represent the flow of electrons during the reaction.

PROBLEM 6-44 Ammonia reacts with acetyl chloride (CH_3COCl) to give acetamide (CH_3CONH_2). Identify the bonds broken and formed in each step of the reaction, and draw curved arrows to represent the flow of electrons in each step.



Acetyl chloride



Acetamide

PROBLEM 6-45 The naturally occurring molecule α-terpineol is biosynthesized by a route that includes the following step:



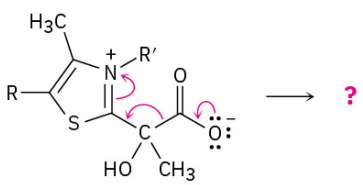
Carbocation **Isomeric carbocation** **α-Terpineol**

(a) Propose a likely structure for the isomeric carbocation intermediate.

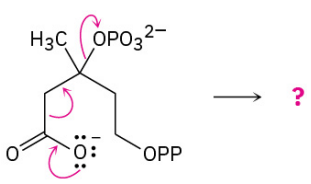
(b) Show the mechanism of each step in the biosynthetic pathway, using curved arrows to indicate electron flow.

PROBLEM 6-46 Predict the product(s) of each of the following biological reactions by interpreting the flow of electrons as indicated by the curved arrows:

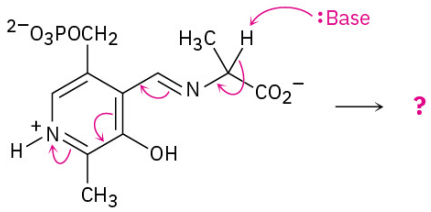
(a)



(b)



(c)



PROBLEM 6-47 Reaction of 2-methylpropene with HBr might, in principle, lead to a mixture of two alkyl bromide addition products. Name them, and draw their structures.

PROBLEM 6-48 Draw the structures of the two carbocation intermediates that might form during the reaction of 2-methylpropene with HBr (Problem 6-47). We'll see in the next chapter that the stability of carbocations depends on the number of alkyl substituents attached to the positively charged carbon—the more alkyl substituents there are, the more stable the cation. Which of the two carbocation intermediates you drew is more stable?